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August 2025

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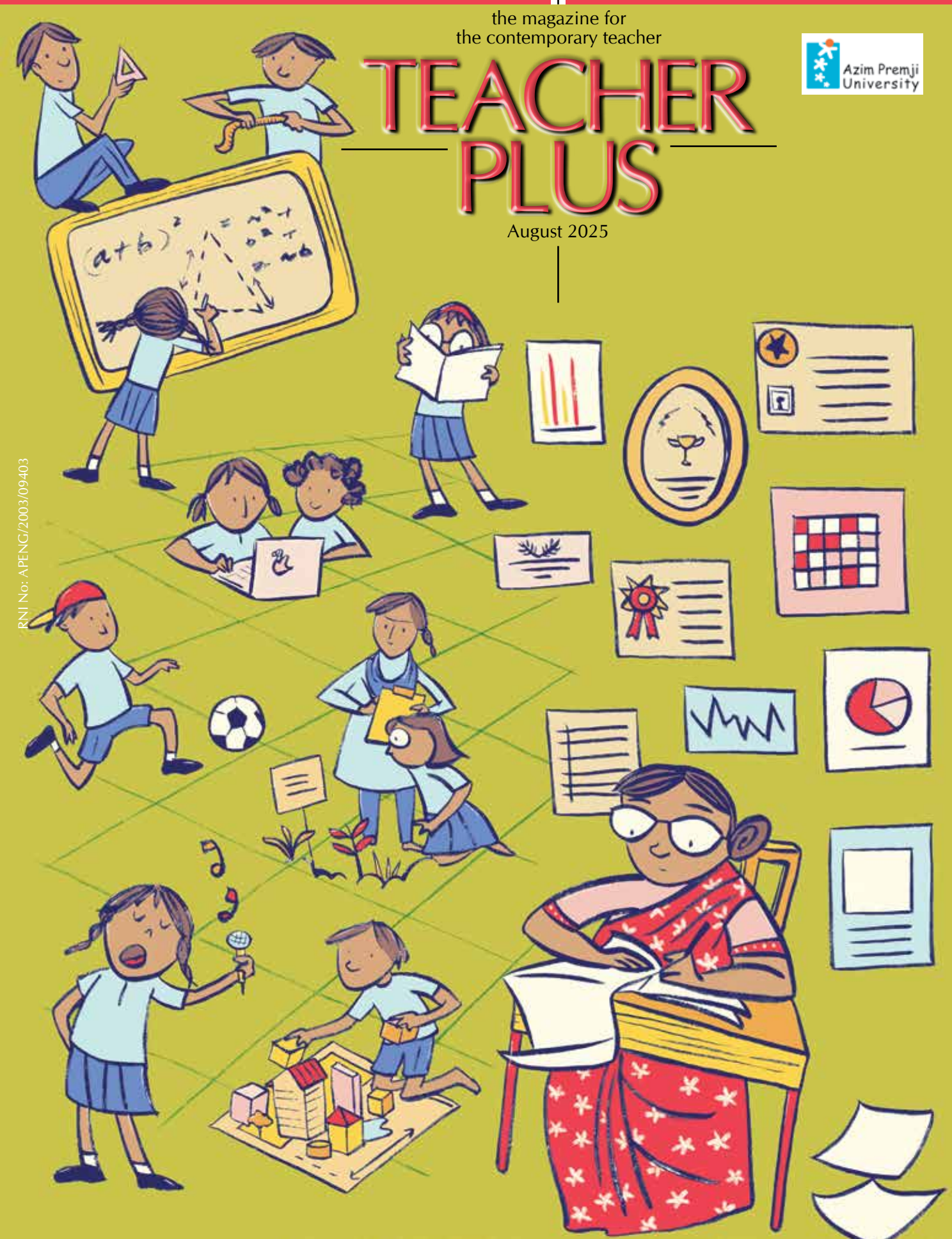

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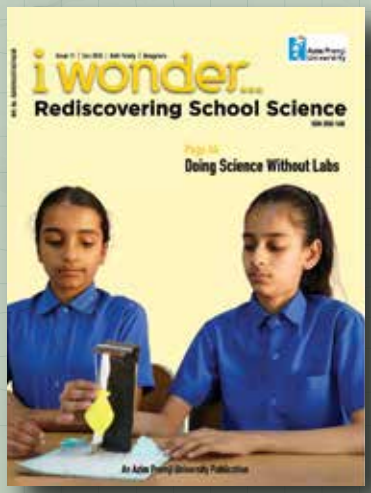

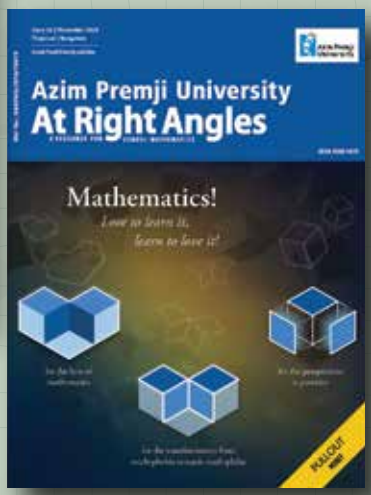

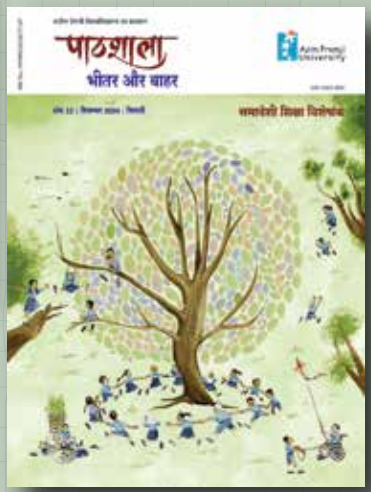

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Cover: Measurement is never simple

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What makes a school safe?

The stream of disturbing incidents in educational institutions refuses to let up, with violence of various kinds that suggests a deep social malaise that cuts across socio-economic categories. Those in charge of policy making and regulation respond in predictable ways, issuing orders that ramp up surveillance and punitive measures. The first takes the form of increasing securitization of schools, while the second calls for stringent action after an incident. The CBSE's latest mandate requires all affiliated schools to install high-resolution CCTV cameras and maintain recordings for at least two weeks. These cameras are to be installed at gates, in corridors, classrooms, libraries, laboratories, playgrounds and other common areas, with washrooms exempted for reasons of privacy. In effect, there will be eyes on members of the school community almost everywhere within the premises.

There's no doubt that surveillance cameras have helped apprehend wrongdoers and clarify the circumstances of a crime, or a troubling incident that may not count technically as a crime. There's also an argument that the knowledge of surveillance can deter potential wrongdoers. This argument also extends to the application of severe punitive measures when crimes are committed – so the presence of the death penalty is considered a major deterrent for what a society considers "heinous crimes". But we know only too well that the existence of stringent laws has not made things safer for vulnerable groups, often because the implementation of those laws is lax, or there are technical loopholes that allow perpetrators to go free. Moreover, the wheels of justice, as they say, move exceedingly slow, and the process can often traumatize the survivor. Surveillance systems, too, can end up being misused to impinge on privacy and freedom – would any of us want to be watched all the time, even if we're not doing anything wrong?

So how do we make our educational spaces safer, and education itself a more nurturing experience? It is partly about having rules and reasonable security measures in place, no doubt. But a sense of safety doesn't come only from sturdy walls and strict rules; it should also be about being safe in other, less tangible ways. Safe to speak our mind. Safe to experiment. Safe to explore and discover. Safe to make mistakes and learn from them. A school's safety agenda should include these protections as well, for all children.

It's a tall order, no doubt. But if schools can't be safe spaces in all these ways, then what place can?

Usha Raman

The betting app craze in India

A challenge for students, teens, and educators

Neerja Singh

In the digital age, India's youth are navigating a landscape transformed by technology, where opportunities and risks coexist in unprecedented ways. Among the most pressing challenges emerging today is the rapid rise of online betting apps – a phenomenon that has swept across the country, captivating millions of young people with the allure of quick money and excitement. What is this betting app craze? What is its impact on students, teenagers, and educators? And why is there an urgent need for awareness, education, and regulation to safeguard the future of India's youth?

The digital gateway to betting began with the explosion of affordable smart phones and cheap internet data that democratized access to online platforms, including betting apps. With just a few taps, young Indians can download apps that allow them to wager on cricket, football, and virtual casino games. These apps boast sleek designs, aggressive marketing, campaigns, and promises of instant financial gains, making them highly appealing to students and teens who often dream of supplementing their income or achieving financial independence quickly. The fantasy of turning a small bet into a life-changing sum is a powerful draw that many find hard to resist.

Cricket, India's most beloved sport, plays a central role in this trend. Major tournaments like the Indian Premier League (IPL) see a surge in betting activity, with millions placing stakes on every ball bowled. Betting apps leverage this passion by employing celebrity endorsements, sometimes even using AI-generated deepfake videos of cricket icons such as MS Dhoni and Virat Kohli to build trust and attract users. Social media influencers and celebrities further amplify the message, often glamorizing betting as a fun and lucrative pastime.

But there is the dark reality of addiction and financial ruin beneath the glitz and glamour. Many young users fall prey to addiction, lured by the psychological tricks embedded in these apps. The thrill of near-wins, intermittent small payouts, and the dopamine rush from betting create a potent psychological trap. Stories abound of students and young professionals who have lost significant sums – sometimes running into lakhs of rupees – leading to financial distress, academic decline, and strained family relationships.

One poignant example is the case of Ashwini, a 17-year-old from Delhi, who became addicted to online betting apps during the pandemic. What began as casual gaming to redeem reward points quickly escalated into gambling for money. Over the course of a year, Ashwini borrowed from friends, pilfered valuables from home, and even accessed his father's bank account to fund his habit, losing over ₹2 lakh. When confronted, he initially denied the addiction, but later admitted to his compulsive behaviour. His parents paid off his debts, hoping he would stop, but he resumed betting and became aggressive when denied money. Ashwini was eventually brought to psychiatric treatment for gambling addiction, highlighting the severe mental health and familial consequences of such compulsions.

Many apps operate in legal gray zones or outright illegality, using fake games and manipulated odds to keep users hooked. When users attempt to withdraw their winnings, they often encounter technical glitches, unresponsive customer support, or app shutdowns, leaving them with losses and no recourse. The consequences extend beyond finances. Mental health issues, including anxiety, depression, and in extreme cases, suicidal tendencies, have been linked to betting addiction among the youth. The stigma around gambling often prevents affected individuals from seeking help, exacerbating their plight.

As to the scale and societal impact, the betting app epidemic is vast and growing. Reports indicate that during major sporting events, nearly 370 million Indians engage in online betting, with annual losses estimated at over ₹1.1 lakh crore. Despite government efforts to ban illegal sites, new platforms

continue to emerge, often operating from outside India and using cryptocurrencies to evade detection. This not only threatens individual finances but also facilitates organized crime and money laundering.

For students and teens, the impact is particularly acute. Peer pressure, easy access to technology, and the glamorization of betting create a perfect storm. Many young people start betting as a way to make sports more exciting or to earn quick money but soon find themselves trapped in a cycle of compulsion and loss. Academic performance suffers, relationships with family and friends deteriorate, and the risk of mental health crises increases.

What then is the role of educators and schools? Teachers and educational institutions are on the frontline of this challenge. Awareness of the risks of online betting must become integral to school and college curricula. Open conversations can demystify the glamorous facade of betting apps and expose the harsh realities faced by many users. Financial literacy programs should equip students with the skills to manage money wisely and recognize the dangers of quick-money schemes.

Image generated using AI.

Schools can also provide safe spaces for students to discuss their concerns and seek help without the fear of judgment. Educators can collaborate with parents and community leaders to launch awareness campaigns that highlight real-life stories and promote resilience against the lure of betting apps.

As to legal and regulatory responses, India's legal framework on betting is complex and fragmented, with enforcement challenges exacerbated by the cross-border nature of many apps and the use of cryptocurrencies. The government has intensified efforts to block illegal sites and investigate scams, but experts argue that stronger regulation, better enforcement, and international cooperation are essential to curb the menace effectively.

Stricter Know Your Customer (KYC) norms, monitoring of digital payments, and swift action against illegal operators can protect vulnerable users. However, laws alone are insufficient without comprehensive public education and victim support systems.

What is also needed are robust support systems for victims, including counselling, financial advice, and legal assistance. Breaking the stigma around gambling addiction and creating accessible help channels can encourage affected individuals to seek aid early, preventing further harm.

The betting app craze is more than a passing trend – it is a digital epidemic that threatens the wellbeing of India's youth. While technology offers immense opportunities, it also poses risks that demand vigilance and responsibility. Students, teens, educators, families, policymakers, and society at large must collaborate to protect the next generation from the seductive but perilous world of online betting.

By fostering awareness, strengthening education, enhancing regulation, and providing compassionate support, we can ensure that India's digital revolution empowers young people rather than ensnares them in cycles of addiction and loss. The future depends on a collective commitment to turning the tide against this growing challenge.

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Assessing change

Trends and transitions in school assessment in India

Simran Luthra

Let me begin with a disclaimer. This article is an ambitious attempt. It does not claim to be comprehensive or definitive. Instead, it aims at being a conversation starter by provoking thought and reflection. It is an attempt to reflect on and discuss the current trends and status of a critical aspect of school education: assessment – a factor that determines the worth and quality of teaching and learning, as well as institutional status. However, given where we stand today, I argue that assessment is a complex, variegated, and uneven phenomenon in India. Hence, rather than offering definitive answers, this article aims to explore current trends, unpack ideas, and make suggestions.

Alright, now that I feel satisfied after that elaborate disclaimer, let us outline some facts about assessment policy reform in India to set the context:

- Policy reform in assessments was first attempted in the National Curriculum Framework (NCF) 2005, which sought to shift pedagogy to a constructivist mode from a rote-based one. NCF 2005 also introduced Continuous and Comprehensive Evaluation (CCE) with the aim of reducing pressure on students and for 'diagnosis, remediation and enhancing of learning' (NCF 2005).

Illustrations: Sunil Chawdiker

- The Right to Education Act (2009) further emphasized CCE and introduced the no-detention policy.
- The National Education Policy 2020 and NCF 2023 emphasize experiential learning, competency-based learning outcomes, and continue to reiterate and emphasize CCE focusing on learning outcomes over rote learning. There is a renewed emphasis on assessment *for*, *of*, and *as* learning.
- In December 2024, the no-detention policy for grades 5-8 was discontinued amid concerns that students were being promoted without achieving the required learning outcomes. The policy change also mandates remedial instruction and re-examination opportunities.

Policy on paper and policy on ground are two very different realities. I, therefore, spoke to teachers – those who are expected to be the agents of policy reform – to get a sense of their experiences and understanding of where assessments stand at present.

Scholars consistently assert how teachers' voices are crucial to understanding ground realities. At the same time, in the Indian research context, teachers' voices remain marginalized (Batra, 2005). Although my approach is journalistic rather than that of rigorous academic research, I believe it is essential to pay attention to what teachers are saying as their insights can both indicate gaps as well as possible solutions.

I spoke to a mix of government school and private school teachers. What emerged was a picture of differential experiences of teaching and assessing. There were region-wise differences, with Delhi teachers reporting high adherence to the latest NEP and NCF recommendations and lower adherence in states such as Kerala, Telangana, and Bihar.

Further, differences in teacher experiences stemmed from two key factors: school type and socio-economic profile. Government schools, with stricter compliance requirements, saw teachers adhering more closely to activity-based and experiential methods due to documentation expectations. Additionally, teachers in resource-rich private schools had vastly different experiences compared to those in low-fee private schools, highlighting the impact of socio-economic disparities. Thus, teachers are able to experiment with constructivist pedagogy, which is activity/project-based with greater ease with students who have support at home and/or internet access.

In what follows, I have summarized insights that emerged from their interviews into themes and/or trends. If you are a teacher or educator, it could be that some of these resonate with your experiences while others do not.

Pressure of documentation

Most teachers shared how the shift in focus from summative to formative assessments has resulted in extensive documentation for them. This includes both paper-pen documentation as well as digital, meaning double the labour. This is a known problem, but one that remains unsolved.

The introduction of the Holistic Progress Card (HPC) as recommended by NEP 2020 and NCF 2023 is not uniform with some schools having implemented it independently and others not. Further, the form in which the HPC is used is also not uniform, as there is no set template for it yet and schools and service-providers have interpreted it differently. However, in instances where it *has* been introduced, based on the suggested format, teachers have found certain sections to be redundant or overkill.

Divya,* for instance, shared how she considered it a waste of time for children to evaluate their peers on academics, which is a field that is part of their HPC document. On the other hand, Sherin reported how the lack of clear guidelines on how to carry out peer evaluation made it hard for teachers to implement it even though she believes it is something that can and should be done. Both Divya and Sherin teach in government schools in Delhi and Kerala respectively.

Sherin also shared how the pressure to enter scores or grades for multiple formative assessments mandatorily and frequently often encouraged teachers to enter marks at random, thus defeating the very purpose of assessing and reducing it to a mere formality rather than a meaningful activity.

Alignment of pedagogy and assessments

I was particularly interested in exploring the alignment between pedagogy and assessments. India has had a strong textbook-centric school culture, and the teachers' own socialization has taken place in the rote and textbook-oriented system, something that Shraboni, a private school teacher from Ranchi, also acknowledged. Shifting from this pre-existing textbook culture to a child-centric one that promotes 'joyful learning' and is activity and project-based has been long coming.

*Names changed to maintain confidentiality of teacher identities.

From a policy perspective, as early as 1966, the Kothari Commission Report and then the New Policy on Education (NPE), 1986, both recommended child-centric approaches. NPE 1986, in fact, first introduced the adjective 'joyful' for school education in India:

> By making elementary education child-centred, we would be introducing a long-awaited reform in the system. The most important aspect of this reform will be to make education a joyful, inventive and satisfying learning activity, rather than a system of rote and cheerless, authoritarian instruction. (NPE, 1986)

However, even after four decades, textbooks remain firmly entrenched in the Indian classroom and policy documents continue to make similar recommendations.

It would, however, be wrong to say that nothing has moved. Most teachers reported a pedagogical shift from a textbook-oriented one to activity-based methods. Driven by policy and supported by training initiatives, activity-based learning is becoming the norm in most schools. This is also supported by teachers' own reflection on the kind of rote-based education they experienced and wanting to provide a more meaningful experience to their students. Thus, pedagogical reform seems to be evident. But, what about assessment reform?

Shifts in assessment seem to be much more uneven despite policy recommendations. Various studies have shown how CCE, despite being a progressive move, did not meet much success (Juneja, 2018; Berry et al, 2020) after its introduction in NCF 2005. The failure is attributed to various reasons, ranging from conceptual differences in policy and implementation frameworks that have created confusion from the outset (Yagnamurthy, 2015), the lack of orientation of teachers towards policy changes in assessments (Brown et al, 2014), and other systemic factors such as classroom size,

focus on syllabus completion, and so on that make implementation realistically difficult. The teachers' narratives that inform this article corroborate most of the existing research, which is, in fact, minimal and scant.

Divya and Shraboni, both lamented that in their respective schools, teachers focused on joyful and activity-based learning throughout the year, but that the rote-based pen-and-paper exams were not in alignment with such a pedagogical approach. Sherin elaborated on this further when she said that at times children who are not the best at expressing themselves through writing but have shown evidence of conceptual understanding through activities, get 'poor results'. The reverse is more common: children who are able to rote learn well, end up with good marks despite lacking conceptual understanding. In such cases, the teachers strongly feel that there is a misalignment between pedagogy and assessments.

'Formative' assessment: now a misnomer?

The purpose of formative assessment is assessment *for* learning, as an input for teachers to adjust instruction or address conceptual or skill gaps. By definition, it is supposed to be low stakes, ongoing, and flexible. However, most teachers reported that 'FAs' are conducted as paper-pen tests of lesser marks (typically 20 or 30) and function as low stakes exams. Instances of formative assessment not being paper-pen based are rare but not nil.

Malati, whose teaching career spans three decades, shared how CCE in the 2010s was substantially different from its current form. Formative assessment consisted of projects or hands-on activities and was true to its definition and purpose. She highlighted how the sudden shift then became overwhelming for teachers and students alike, with all subject teachers assigning multiple projects throughout the year, leading ultimately to rejection of the CCE model nearly a decade ago. However, now the 'FA' component is equal to what used to be called 'unit tests' earlier; almost defeating the purpose of it being 'formative' in nature. In Malati's school, which is a private school in Delhi, the teachers and management have struck a middle-ground where they have gone back to implementing CCE in its original spirit but ensuring they do not overburden students or teachers with too many projects.

English teachers shared that while their classroom activities focused on speaking exercises along with reading and writing, a holistic approach that catered to the four skills of Listening, Speaking, Reading, and Writing (LSRW) was sorely missing in English assessments at all levels from primary to high school with overemphasis on reading and writing. Only in very limited instances did teachers report that formative assessments were built around debates or speaking tasks. This is a major concern because it means that language continues to be taught as a 'subject' as opposed to as a set of competencies.

Are higher-order or critical thinking skills being tested?

When asked if the question papers (summative and/or formative) contained questions pertaining to critical or higher-order thinking skills, most teachers said they did not. Only a few said that one or two such questions were included, but overall the papers were rote-based.

Further, it is mostly in private schools with children from resource-rich families, where higher-order or critical thinking skills were a focus in assessments. Students in schools belonging to the opposite end of the spectrum struggle owing to low reading levels.

The role of foundational literacy in assessments

According to teachers, a major impediment when answering pen-and-paper exams was language proficiency. Low reading and writing levels seem to be responsible for unsatisfactory exam performance. Foundational literacy (along with numeracy) has been rightly identified as a gap by NEP 2020 and NCF 2023, which strongly recommend the development of these basic skills, particularly by grade 3, so that students can become independent learners at the earliest. However, the actual realization of this is not as straightforward owing to language being taught more as a 'subject' than as a set of skills as mentioned earlier.

Parents and assessments

While pedagogy and curriculum are aspects that parents in mainstream Indian schools hardly get into, when it comes to assessments that is not the case. Parent-Teacher Meetings are typically 'report-card days', where results from formative and summative assessments are shared with parents and hence become high stakes.

Some teachers reported that the recent shift from marks to grades was something that several parents were not used to and how they still demand percentages and ranks, as problematic as these measures are. This indicates how culturally

embedded schooling practices become and how it is difficult to pivot from them.

At the same time, sudden policy changes can cause confusion. The scrapping of the no-detention policy is a case in point. Divya noted that students who have never faced detention due to exam performance are now being caught off guard. As a result, teachers are having to counsel both parents and students through this transition.

On a more hopeful note, a positive generational shift, reported by Divya, was that younger parents seem to be less concerned with marks and grades. A significant chunk of parents of younger children are millennials who are more keen on holistic development and eager to protect their children's mental health from the tyranny of marks and grades.

Some key lessons...

Here are some vital learnings based on the teachers' narratives.

Everyone can do something in their context: Teachers continue to innovate despite constraints. Priya, a headmistress and teacher from a school in Hyderabad, shared that students in her school have limited internet access and minimal academic support at home. To adapt, teachers design project-based formative assessments that require extra effort but are necessary given the students' socio-economic backgrounds. Efforts are made to ensure that students relate concepts with their daily lives: in maths, students create word problems based on daily life, while social studies projects focus on current affairs, with teachers providing additional support. Such approaches reflect teachers' dedication to making learning relevant and engaging despite challenges.

NEP/NCF options to be available across states: NCF 2023 allows students to drop traditional subjects and take up vocational subjects such as music, information technology, or entrepreneurship from grade 9 onwards. Malati shared how this was being implemented in her school for the last couple of years and has been received very well by students.

It was interesting to hear Priya raise this same concern. She shared how academics was not every student's cup of tea and that vocational options should be available to students. Hence, while vocational options are available to students in Delhi, they are not available to students across the country, particularly in state-board schools. This is something to be considered from a policy perspective.

Classroom size to be contained: Most teachers shared how their class size was 40 on average. Malati said that in her career of nearly three decades, she had witnessed a steady rise in the number of students per class. However, as she mentioned, this has an adverse effect on the connection a teacher is required to make with students and the individual attention each child needs. Meaningful and regular formative assessments also get compromised if the class size is too large. School leaders and management need to address this issue if they aim to provide quality education.

The bane of 'syllabus completion': One of the remnants of a textbook-centric culture is the expectation that the entire book must be 'covered' by the teacher from end to end. However, when 'syllabus coverage' takes priority over depth and conceptual clarity, it becomes counterproductive. Further, if assessments remain rote-oriented, it puts even greater pressure on students. Schools and teachers should realistically and critically rationalize the syllabus and greater autonomy should be given to teachers to make such decisions.

Leveraging technology to make teachers' lives easier: Authorities at both state and school levels should take steps to eliminate redundant documentation for teachers. Timely feedback shared by teachers on the items to be documented can help reduce redundancies or misaligned fields to be filled which can both save time and effort and also avoid erroneous or low-quality data. Again for this, we need to *listen* to teachers.

To conclude...

As I conclude, all that remains unsaid weighs heavy on me. There is so much more that the teachers shared that remain outside of my word count limitations for this article. And I would reiterate here that the intention of this article is to get us thinking more and more critically about assessment practices.

I originally set out to write this article to explore the complex relationship between pedagogy and assessments. We know from anecdotal records and considerable research that assessments impact pedagogy; the 'teach to test' phenomenon, for instance, in which teachers tend to teach in a manner that aligns with how students will fare in exams at the expense of deeper learning or broader curricular goals (Phelps, 2011).

There is ample research evidence that summative assessments affect the pedagogic practices of teachers (Erskine, 2014). With standardized testing becoming the global norm, researchers have found increasingly, a narrowing of curricula taking place, with a growing focus on testable content where memorization takes priority over understanding to achieve high scores. It has also been found that this leads to examination subjects gaining a higher status than non-examinable subjects such as physical education, art and craft, and music. The teacher narratives echo the patterns observed in these wider studies. Assessments hence deserve the sort of focus and effort dedicated to pedagogy.

It is important to clarify that I entered the 'field', so to speak, with my own hypothesis or biases. I thought most teachers I would speak with would be despondent or that there would be considerable confusion. Teachers, however, are an optimistic lot and a wealth of ideas, perspectives, and energy reside in them. Greater collaboration and knowledge sharing among educators, coupled with consistent dialogue and rigorous research, are essential for working towards meaningful and effective assessment practices and policies.

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Tackling student misbehaviour

Tenzing Rapgyal

We are a school in the peaceful foothills of the Himalayas, but the year 2023 was anything but peaceful for us. The students of the 12th standard that year became notorious for being the most difficult batch to deal with. The school environment was also filled with negativity caused by a series of unfortunate incidents:

- An acrimonious altercation between some senior students and members of the advisory committee on their hostel rounds during an evening self-study hour.
- The public display of conflict between a teacher and a student during the morning assembly.
- A theft in the school tuck shop by middle school students.
- A viral case of some 10th class students caught on camera in an inebriated condition in their hostel.
- An account of a student with a violent past running after the foster mother in the hostel.

What are the causes of such misbehaviour?

Corporal punishment was banned in our school in the early 2000s, making disciplining students a challenge for the teachers. Children, therefore, are free from the fear of punishments. Another challenge for the teachers is to make their classes innovative and interesting even as they race to finish the syllabus and get students exam-ready. While students are required to write internal assessments as well, they add only 20% to a student's final result. It is the year-end final exam that makes up 80% of student assessment. This means that passing the final exam is much easier. All that a student has to do is to study intensively for a month or two before the final exam. Such a system seems to have been designed to help students pass, so that they don't drop out of school. So, unless pushed by parents at home, students have neither intrinsic nor extrinsic motivation to study, leading to disciplinary problems in class and hostels. Trying to

change this systemic failure is like hitting one's head against a wall. But, that does not mean one should give up and surrender to reality. By being wise and realistic, one can deal with the situation, which is what I have been trying to do after falling prey to hopelessness myself.

What are the types of disciplinary problems in my class?

The common disciplinary hindrances that I have had to deal with are bunking, being late after break, sleeping even during the first period, irregular class work, incomplete home assignments, talking in class, not bringing the textbook, or being disruptive. Among these, it is being late to class that irritates me the most, because it disrupts the flow of the lesson. To deal with this nuisance, I have tried using humour. For instance, when students would knock on the classroom door while the lesson was going on, I would stop teaching and let them enter to a thunderous applause of the students already in the class. But, the punctual students were losing precious time because of this. Therefore, I made a rule that if students are late, but find the classroom door ajar, they should enter silently without asking for permission. If, on the other hand, the door is bolted from inside, the students will be considered absent, as they have been late for more than 20 minutes of the period.

What remedial measures I have taken

Over the last five years, I have devised a system of Six Levels of Remedial Measures. Earlier, my remedial steps were reactive because I did not have a system to deal with repeated lapses. There was no clarity about the consequences of a breach. As a result, often, there were verbal altercations between the problematic students and me. To avoid such confrontations, I came up with the following system, and so far the results have been satisfactory.

Level	Remedial Measures
1	I. Three chances to improve. II. If a student does not show any improvement even after the third chance, there will be a one-on-one talk with the student. III. In the case of being absent, the student will be reported to the headmaster's office every time they are absent and will be detained during break/after class/weekends to do manual work.
2	I. If a student has already been through Level 1 and yet failed to improve, he/she will go through the first and second steps of Level 1, except in the case of being absent. II. If a student still does not show any improvement, he/she will be detained to do community service on holidays.
3	I. If there is no improvement after Levels 1 and 2, the student will be referred to the headmaster's office for further remedial actions.
4	I. If the problem continues, the student will be referred to the headmaster's office for a meeting with the concerned teacher, their parents/guardians/warden/matron for further remedial actions.
5	I. If there is no improvement even at this stage, the student will be referred to the principal's office for further remedial actions.
6	I. At this last stage, the student will be referred to the headmaster's office for an urgent meeting of all the concerned stakeholders, i.e., the principal, the headmaster, the teacher and the student's parent/guardian/warden/matron for further remedial actions.

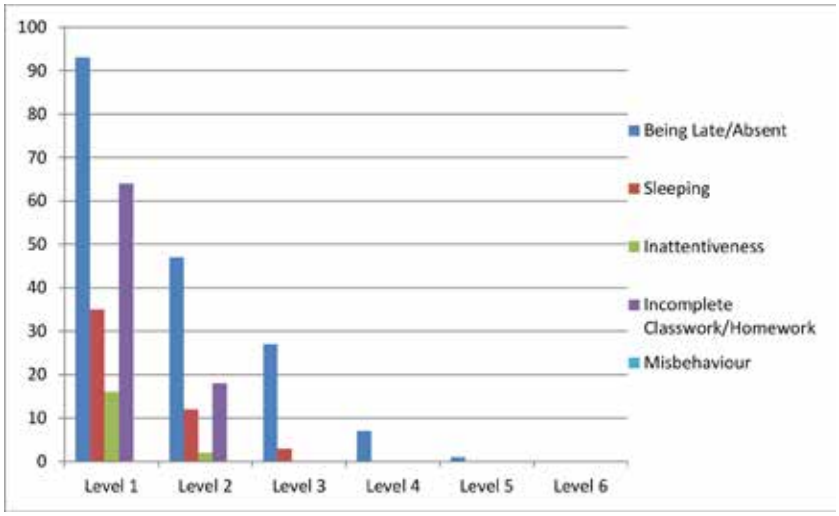
How effective have the remedial measures been?

To answer the question, let me present a quantified response. Since the beginning of the session in March 2023 to the end of January 2024, I had been keeping a record of the lapses on the part of the students, which number to 1049.

Tabulation of the number of cases

Levels	Being Late/ Absent	Sleeping	Inatten-tiveness	Incomplete Class work/ Homework	Misbehaviour
1	93	35	16	64	0
2	47	12	2	18	0
3	27	3	0	1	0
4	7	0	0	0	0
5	1	0	0	0	0
6	0	0	0	0	0

Graphical representation of the data



It is evident from the data that there was a steady decline in the number of cases across the different categories of lapses.

For example, the number of cases of being late/absent in Level 1 dropped from 93 to only one by Level 5. Sleeping students went down from 35 in Level 1 to three in Level 3, after which all students rectified this error. In the case of inattentiveness, out of 16 students in Level 1, only two crossed into Level 2. Regarding the regularity of students in their daily class work, 64 cases were recorded in Level 1, which decreased to merely one in Level 3.

Only in the case of 'being late/absent', seven students reached Level 4, which means that their parents/guardians were contacted to report the matter and requested to guide their wards and only one student was reported to the principal's office for crossing over to Level 5. In the rest of the cases, after Level 2, which involves detention, there was a drastic decline with only three students being reported for sleeping to the headmaster's office for crossing Level 2 and one student for his/her incomplete class work.

Were my remedial measures effective?

I think so. However, the cause for concern was the alarming number of cases of 'being late/absent' in Level 1, which numbered to 93 students out of 171 students of classes XI and XII. In this category, the decline in the case is almost half at each level till Level 4. What the data suggests is that again almost half of the total number of students took undue advantage of the three chances/warnings given in Level 1. Most of the cases in this category are students being late to class after the short recess and lunch break. In other categories of lapses, the breach in Level 1 is higher compared to the subsequent levels. My intention behind giving students three chances/warnings is to provide them opportunities to self-discipline without extrinsic remedial measures. But on studying the data, the question is whether I should do away with the three chances of intrinsic remedy. One encouraging aspect the data reveals is that students restrain from crossing Level 3, which can be deduced as fear or respect for their parents, which deter them from being reported.

Finally, what I have learned from implementing this system for three years is that if there are clearly defined consequences for a breach of classroom decorum, students regulate their behaviour. Some students may misuse the remedial opportunities given to them, but knowing their stories (of broken families, dysfunctional parents, death of a parent, and so on), I realize that I have to be sensitive to minor lapses when their internal self is in turmoil.

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Connecting with language

Ashish Oommen John
(with inputs from Preeti Jangra)

Namaste, Sofia. Namaste, Niya. Namaste Arya.
Namaste.
Namaste.
Kya khaya?
Doodh.
Anda.
Bread.

This is how Preeti's classes with Arya, Niya, and Sofia would begin. Learning another language can often appear daunting – different rules, different ideas, different ways of talking about something. Language isn't just something you spend countless hours learning in school in the hope of scraping through an exam. Language is so much more – it has history, culture, tradition, innovation, and emotion. It is also informed by our unique and varied experiences. Language is power and belonging. But, of course, when little children have been asked by their parents to take after-school Hindi lessons, they don't care about any of this.

Let's meet the children. Arya is the oldest. When Preeti began working with her, she was nine years old and already had a basic vocabulary and understanding of Hindi. Sofia and Niya, both seven years old, did not speak the language at all. Preeti meets them online once a week for 45 minutes. The children are based in the UK, and while they do encounter Hindi on their visits to India and occasionally through family and friends, they do not have the need to engage with it on a regular basis. As the one with the most enhanced Hindi-speaking ability, Arya routinely lords over her sisters by conversing solemnly with her father in this secret tongue – asking for important things like more butter on her toast.

Play

Preeti's methodology consists of three elements that blend together – play, conversation, and connection. Each of these will be detailed in this article. We begin with play. As adults, we can sometimes have a sceptical view of play. It can be seen as a waste of time, or at best, as something that may be tolerated, or on occasion, encouraged as a *hook* in classes, after which we must get down to business. Perhaps, it is the noise that children generate while playing, or just raw jealousy, but we would much rather they didn't.

But we know that children (and adults) learn a great deal through play. Interpersonal skills, inventiveness, confidence, self-regulation – the list can go on. It is also fun. It takes away the seriousness attributed to learning without impacting the child's ability to learn. And, of course, it helps them stay interested in what Preeti is trying to teach them.

Preeti uses a wide range of ways to play while teaching Hindi such as creating a story with the children taking turns to add one sentence at a time or dancing to songs in Hindi and learning to count through it. Sometimes, she asks them questions in Hindi and gets the children to draw their responses. The questions could be as simple as, "which is the fastest animal?" or more nuanced as, "if you know that your parents are going to agree to something, what would you ask for?" By encouraging them to draw, she is able to elicit curious responses. To the second question, for example, the children drew computers or electronic devices of some sort. However, to a question about a memory they have of rain, they drew pictures of the food they had eaten while it was raining outside.

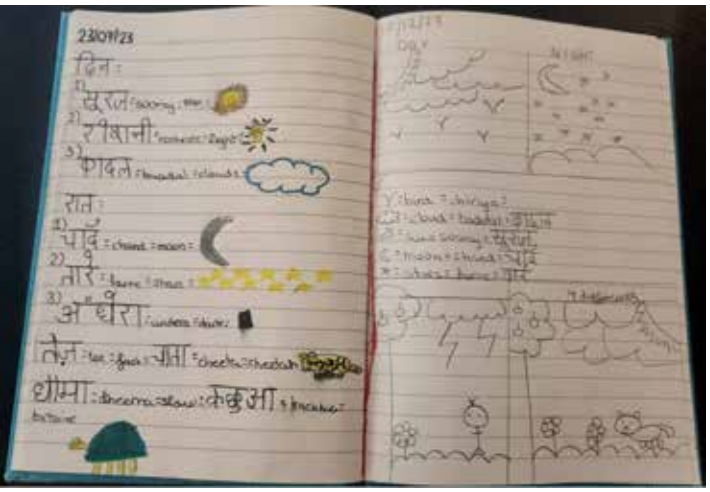
Photos courtesy: Preeti Jangra

The challenges of learning a language are many, but when combined with the fact that Preeti meets the children for one hour per week online, keeping their attention for the entire time is vital. Play interests them enough to continue to be in the class.

Another fan favourite, especially in the early days, was the scavenger hunt. In order to teach them the names of colours, Preeti would name a coulour and the children would scurry off to find things of that colour. It is a wonderful way of ensuring that they don't have to sit still and stare at a screen for the entire lesson while also listening, translating the word into a language they understand, and then hurrying off to act on this knowledge. This aspect of play, however, would be incomplete without the chance to discuss what they had just done through conversation.

Conversation

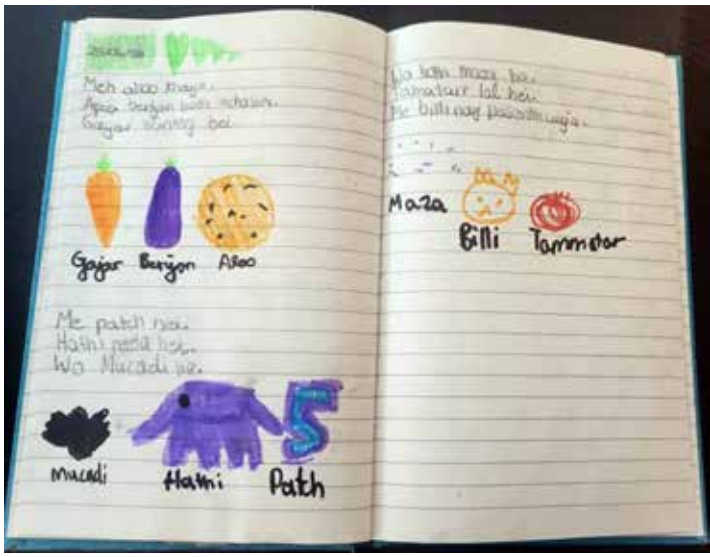
The parents of the children had asked Preeti to focus on listening and speaking skills, and conversations are a fantastic way of doing that. Talking to other people is arguably the fundamental use of language. As human beings, we enjoy talking. Conversations afford us a space within which we can use language in creative ways. Whether it is mispronouncing a word in order to make a joke, or using a particular phrase to invoke thought, or revealing fewer details in order to generate curiosity, we instinctively use language to make our lives richer. Language is most useful to us when we make it our own, when we can play with it, and infuse it with a part of who we are.



These conversations that Preeti has with the children are intrinsically linked to the aspect of play. Both play and conversation are intertwined and ensure that everybody is participating and also using Hindi to express their thoughts. These activities also come with degrees of nuance. At one level, Preeti asks the children questions and they take time to discuss the answers. Further along the complexity scale, they take stories in Hindi and role-play them. Sometimes, after they have learnt the meaning of a particular word or phrase, the children are asked to try and use it outside of the class. They can do this either in conversation with one another or with their parents.

Preeti also introduced the trio to the Hindi movie *Lagaan*. Even one movie can make a big difference. I think back to my schooling when I was attempting to learn French. I thought I had a fairly decent command over the language. And then, we watched a movie. Three minutes in, while simultaneously overwhelmed by the language I clearly knew nothing about and also deeply grateful to the inventor of subtitles, a distinct realization swept over me. The people of France did not speak as slowly and clearly as my teacher. For Arya, Sofia, and Niya, watching *Lagaan*, exposed them to a variety of accents, speech patterns, words, and phrases that Preeti would never have been able to use in class. Through one movie, their overall concept of the Hindi language expanded.

The girls learn a great deal through rhymes as well. These rhymes, about vegetables and animals, add to their vocabulary and allow them access to a large number of words with which they can play. *Nani Teri Morni, Bandar Mama, Lakdi ki Kathi*, and the letter-specific rhymes from Chandrika Mathur's Hindi ki Duniya are some in particular that are used. The students enjoy adding gestures and actions to the rhymes as well, allowing them to engage their entire bodies in the learning of a language. Letter-specific rhymes emphasize one letter of the alphabet by using the sound repeatedly in a variety of words.



N se nana nani pyare,
rehte hai wo nadi kinare,
roj naav ki serrr karate,
neeli nadi ke paar le jaate.
– Hindi ki Duniya, Chandrika Mathur

Another way that Preeti engages the children in conversation is by setting up a pretend shop. Everybody takes turns to be both the shopkeeper and customer. The customer enters the shop to buy something. A conversation ensues, perhaps even a negotiation over the price, with one party attempting to get the better of the other. Finally, there is a resolution in one direction and the customer leaves. While this forms the basic format of the encounter, the variables are different every time, and this helps keep it entertaining for the children. The children enjoy putting on accents, doing funny voices, and coming up with the strangest things to ask shopkeepers. And it is here that the beauty of this game lies. Unlike some other games or activities in which the rules must be followed strictly, this is a free-thinking game. This game is only improved by everybody being silly. And when the children are so focused on being silly and having a good time, they are no longer conscious of making mistakes. It is at this moment that they are no longer worried about learning Hindi; instead they are using it to achieve a different end. The wonderful thing is the level of comfort they feel with the language when they stop trying so hard.

It is in conversation that we are able to better understand another person's ideas, and also our own. It is through conversation that we can truly connect.

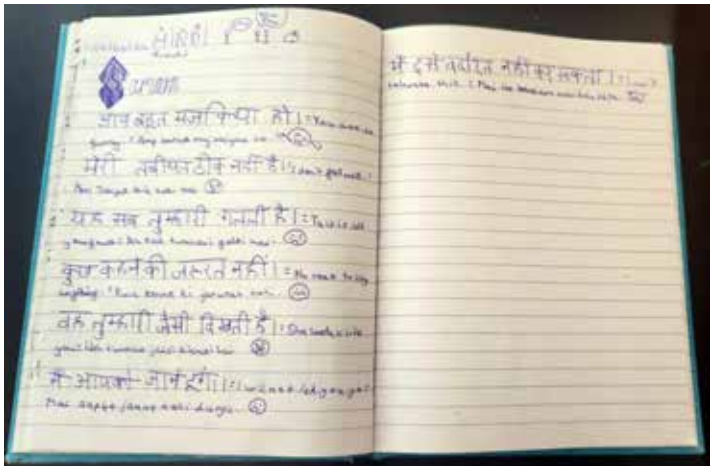
Connection

This brings us to the final element of Preeti's methodology. Connection. What is language if not an attempt at connection? We all seek to connect with other people, with ideas, and with ourselves, and language is what enables it. The teaching of language must account for connection.

Any educator will tell you that it is through connecting with one's students that one is truly able to teach or learn anything. Without that connection, any teaching-learning process is ultimately futile. This is not to say that a teacher cannot teach and a student cannot learn if they don't connect. It certainly is possible. Children do not form deep connections with all their teachers, and vice-versa. However, where there is a relationship based on mutual trust and respect, the learning process is made a great deal easier.

There are few things that can be more embarrassing than learning a new language. We stutter, we flounder, we search for words and sounds that are always just out of reach as all notions of self-respect and suave demeanour abandon us. And yet, making mistakes and having the occasional laugh at ourselves is a faster way to develop linguistic ability than the alternative. The role that the teacher plays here, then, is crucial. If a learner has developed a relationship of trust with their teacher, they know that they won't be ridiculed, singled out, or laughed at for making a mistake. A deep bond between the teacher and learner will go a long way to ensuring that the latter has the freedom to explore, make mistakes, and explore again.

Languages also help us connect with ideas. If a language is a whole world, then learning one is learning to interact with that world. And in interacting with this new world, we gain the tools to reflect on our own – to look at what we already know so well through fresh eyes.



However, one of the most beautiful aspects of learning a language is the ability to find a connection between oneself and the cultures associated with the language. Preeti enjoys engaging her students in discussion about what makes a good teacher, asking them to help her improve as an instructor as well as a human being. This, in turn, helps to ensure that the bond they share keeps growing. Sharing can also bring us closer to strangers at an airport or on a train simply because we know a little bit of their language. Though they might not know it yet, Arya, Sofia, and Niya have embarked on a journey that will afford

them a sense of belonging to a community that they might not otherwise have known.

Namaste, Sofia. Namaste, Niya. Namaste, Arya.
Namaste.
Namaste.
Kya khaya?
Maine doodh piya.
Maine anda khaya.
Maine bread aur butter khaya.

Preeti's classes still begin with *namastes* and asking about how they are and what they have eaten. Now, they respond in sentences and phrases instead of words. They no longer hesitate. While this might not seem like much, there is a world of difference that has taken place. They have played, chatted, and learnt together. And most of all, they have found ways to connect.

**The names of the children have been changed to preserve their privacy.*
**Preeti's work with three children, over the course of three plus years, has shaped our understanding of how to teach a language to children for whom it is foreign. With Hindi, this applies to children based in other countries (as these three children are) just as much as it does to those in other states of India where it is not widely spoken. We wanted to write about the things that we have learnt so that it may be of use to others undertaking a similar challenge. A part of Preeti's system comes from the thinking and writing of Chandrika Mathur – in particular, her curriculum called Hindi ki Duniya, and we are grateful to her for it.

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A simple

In every school, there are students who love solving problems, students who hesitate to try, and students who just pass by without noticing. But something magical happens when a puzzle appears on the whiteboard near the entrance gate of our school.

Every week, I put up a simple yet thought-provoking math puzzle. It is not something from a textbook. It is a reasoning-based question that even primary students can solve if they think hard enough. And then, I step back and watch.

Photos courtesy: Mokhtar Zaman

At first, there is curiosity. A few students stop, tilt their heads, and read the question carefully. Some whisper guesses to their friends, while others pull out their notebooks and write it down to solve later. I have seen students from grades 2 and 3, unsure of the question, discuss it with students of grades 8 or 9, and ask for help. And the best part is that the senior students don't just give the answer away, they guide the younger students, step by step, explaining how they can think about it.

By lunchtime, the puzzle becomes a topic of discussion. Some students confidently claim they have solved it, while others wait to check their answers before submitting. A locked box in front of the staffroom awaits their responses. It's funny how they treat this simple activity as if it were

math puzzle, a big impact

Mokhtar Zaman

a serious mission. They tear small scraps from their notebooks, scribble their answers, fold the papers carefully, and drop them into the box.

A week later, it's time for the big reveal. I unlock the box and go through the answers – some right, some wrong, but all filled with effort. As I separate the correct answers, I imagine the students' eager faces, waiting to see if their name will be displayed on the board.

And then, the best moment arrives. Pictures of the students who got the answer right go up on the board. You should see their reactions!

They rush to the board, calling their friends, pointing at their names. Even those who didn't get it right this time look determined, already asking when the next puzzle will be up.

Why this matters

This isn't just about math. It's about excitement, curiosity, and the joy of learning.

1. **It makes students think.** Instead of memorizing formulas, they learn to approach problems logically, step by step.
2. **It creates a culture of learning.** Younger students get guidance from seniors and discussions happen outside the classroom, making learning natural.
3. **It builds patience.** The waiting period before the results are revealed teaches students that answers don't always come instantly.
4. **It gives a sense of achievement.** When their names with their photos go up on the board, their confidence grows. They feel proud, motivated, and ready for the next challenge.
5. **It makes school exciting.** Instead of just following a routine, students look forward to something unexpected, something that sparks their curiosity.

Week after week, a new puzzle appears, and the cycle begins. And just like that, math stops being a subject – it becomes a part of their everyday excitement, a small challenge that makes their school days a little more interesting.

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PROJECT

It's all in the mud

Meenakshi Umesh

Photos courtesy: Meenakshi Umesh

The natural learning process of every child involves curiosity as the fundamental drive. If children are born to learn, what is the role of a learning space or school? This was the challenge I faced in 2000 when I decided to homeschool my children. And I responded by trusting the process. We can create anything tangible by using the resources of the planet and the efforts of our body. The involvement of the body in the process of meaningful creation of usable resources without the by product of waste is to live sustainably and serve the ecosystem too! The simplicity of life dawned on us. And the richness in that simplicity baffled us! Why then are we all so stressed about our children's education? Because we have complicated life!

At Puvidham, creation is a way of life. We are all involved in physical work to satisfy our basic needs. At one point, the 5th and 6th level children felt the need for a playhouse, and the whole school involved itself in its construction. The knowledge we gained in the process was immense. While it might seem like all play, we encountered math, science, life skills and more in the one month we took to build the playhouse. I understand not all schools have the luxury of space, but I hope this project will encourage teachers to find ways in which their children can work with mud.

Discussion and free expression of thought (language and math)

1. **Establishing the need:** We established a need to learn to work with mud as a construction material and offered the possibility of creating some space for the children's use. Why should we build with mud? What other materials can be used? Why should we build it ourselves?

2. **Deciding the size:** The next step was to decide how big this space would be. Since at 10 years, children do not have an idea of measurement in feet or meters, we proposed that they decide how many children should be able to sit inside the playhouse at any given time.

3. **Arriving at areas:** Then the children had to measure the space needed by one person to sit, stand, and move a little bit. They were asked to outline the space needed for one person to sit comfortably, turn freely, and walk around. They split into groups of five and used chalk to mark on the floor the space required by each one in the group to sit, turn, and walk around. We made squares of one foot by one foot on a chart paper and used them to determine how many square feet of area was needed.

Measurement and calculations of material and space (math, science, and sustainability)

1. **Arriving at a size:** Each group learnt how to use a measuring tape. On an average, it was decided that 5 feet x 5 feet space was needed for each person inside the playhouse, so the playhouse was now a big house of 25x5 = 125 square feet.
2. **Affordability and moderation:** At this time, we sat down to calculate how much money it would cost us to build a space this size. What would the materials be? How will we do the roofing? How can we minimize the size to reduce the cost? After a lot of discussion, we arrived at 10 feet by 8 feet space.
3. **Materials and form:** We decided to build the playhouse with waste bricks and mud as both are almost free of cost. To make an enclosed space with mud, we decided to make a dome so that we wouldn't need wood, which can both be attacked by termites and lead to cost escalation.

Visualizing and finalizing the shape (geometry and design)

1. **Drawing the form and understanding scale:** A rectangle was drawn on a graph paper. The children wanted 10 feet length and arrived at 8 feet width to minimize the space. When the wall thickness of 9 inches was drawn, the children realized that the area was increasing. During discussion, it became apparent that corners are useless, as no one will sit there. So, to further reduce the area, they decided to round off the corners. Since the width was 8 feet, a radius of 4 feet was used to draw two semicircles on both ends of the rectangle. They found out the area of the space by counting the squares on the graph paper. (The formula for calculating the area of a circle was used and the areas were cross-checked.)
2. **Finalizing the form:** This capsule shape we felt could easily be constructed with bricks and mud. Using the centre of the two semicircles we could construct a semicircular dome and join it at the top to make the roof. But this would mean that the height of the playhouse would only be 5 feet.
3. **Ergonomics and entrance design:** The groups then did a study of the heights of the children in their class and agreed to a door height of 3.5 feet. Though most of them were around 4 feet 6 inches tall, they felt that they could bend a little to enter the play space and decided to build an arched doorway. A circular opening on the opposite side would act as a window.

Clearing the site

Digging the foundation

Stone - dry rubble foundation

First layer of bricks

Preparing the soil for mortar

Transporting materials by wheel barrow

Using geometry to create form

Bringing our drawings to life (math)

1. **Making a complete set of drawings:** After making the drawing on the graph paper, we had to learn how to draw using geometric tools and understand the geometry of the structure. Each child had to make the drawings separately using their geometry sets. They started by learning how to draw perfect right angles, how to ensure that the rectangle is perfect, how to mark the centre of the semicircle and then how to draw it, and how to mark the door and window in the plan. They also learnt how to make the elevations and understood how we would construct it. The walls were straight for 1.5 feet and then they curved inward to close into a dome. The drawing took three days of working two hours every day before everyone was able to draw the plan, sections, and elevations.
2. **Calculating the number of bricks needed:** To calculate the number of bricks needed, we had to know the perimeter of the structure we were building. So, each group tried to get an idea of the perimeter with the help of the 7th and 8th class mentors. We had to help them understand the surface area of the playhouse and the surface area of one brick with mud mortar. The division was done and the number of bricks was arrived at.

Getting down to work (language, geography, and math)

1. **Deciding the location of the play space:** The children wanted to have the play space away from the classrooms. But the adults expressed concern that they would not be able to keep an eye on the children. The children wanted no supervision and the adults were concerned about safety. It was agreed that if for one week we had no complaints about the 5th and 6th class children, then we could tuck away the play space, but if any one made even a single complaint against them, the play space would be closer to the classrooms. Even before we started to build the playhouse, however, there were issues about who could use the playhouse and when. So finally, it was agreed that the playhouse had to be in the vicinity of the classrooms, so that the adults could intervene in case of a serious issue.
2. **Marking on the site:** When the children were drawing the plan on the graph paper, it was relatively easy. But marking the structure on land needed a different set of tools. Using twine and golden triangle to mark the rectangle and then the centres of the semicircles and use a twine and stick as a compass to mark the curve was a very interesting transition.

Testing the mud (geography and science)

1. **Sedimentation test:** We filled 10 cms of the mud in a transparent bottle and mixed it with water till it dissolved. We left it to settle overnight. The mud settled in layers. The mud profile was drawn by the children and the percentage of the sediment was determined by measuring the thickness of each layer. We found that the mud had three percent clay.
2. **Mud cake test:** The children took some mud in one hand, pressed it down with the other, and turned their palms upside

down. Depending on the time the mud cake took to fall off, the tensile strength of the mud was determined.

3. **Cylinder roll test:** The mud mixture was rolled into a long cylinder. The longer the cylinder, the more the tensile strength.
4. **Ball test:** Two mud balls of the same size were made and left to dry. One in the shade and the other in the sun. The cracks on the ball told us about the bonding property of the mortar.
5. **Impact test:** The balls were then dropped from a height of 10-12 feet to check the brittleness of the mud and how it would perform under compression.

Improving mud quality (biology and microbiology)

1. **The ratio of mud, sand, and water:** Mud, sand, and water were mixed. The children saw how the mud felt and then covered it and left it to ferment for 24 hours. The next day they touched the mud mixture again and saw the difference in pliability.
2. **Fermentation and microbes:** Why do people use fermented mud for construction? It improves the plasticity and binding properties due to proper soaking of all particles in the mud. Mycelium, a fungus that is there in all kinds of mud grows and spreads through the mud mixture and increases its binding property.
3. **Plants used to improve the plasticity of the mud:** Plants like aloevera, neem, tulsi, milk weed, nirgundi, etc., were collected, crushed, and added to a drum of water. One kilogram of each of the leaves was used.
4. **Kitchen herbs:** Three kilograms of Kadukkai seeds were pounded and made into a powder. This was added to 200 litres of water in a plastic drum. Five kilograms of Jaggery was added to this. A quarter of a kilogram of turmeric powder, ½ kilogram of fenugreek seed powder, and one kilogram maida flour made into a gruel with 10 litres of water were all added to a drum. This mixture was stirred everyday for one week to ferment it. This water was used to turn the mud kept ready

Making your own mud constructions

This type of a structure can even be built on a terrace for children as a space to read or play. Moving materials can be fun when we need to use physics of simple machines to pull the materials up using pulleys. A tour for collecting plants and herbs to mix with the mud will increase children's sense of wonder at the contribution of plants in traditional construction with mud.

The topic of mud construction can be connected with many subjects.

Geography: Soil formation, soil types, rock types, soil colours, topography of India.

History: History of early man and the ancient civilizations and old structures.

Science: Fermentation of the soil and the role of mycelium in increasing the strength of the soil. Soil structure and content, soil Ph, etc. Use of machines to transport material, work and effort, and pulleys and inclined plane.

Math: Area of all shapes, dimensions of shapes and their perimeter, volume of material needed to build, calculation of number of bricks, stones, etc.

Language (mother tongue): Children can document this project in their journals. At Puvidham, our children wrote in their mother tongue about each day –

what they did, what they understood, and what tomorrow might bring.

They wrote of the sun on their backs, the weight in their arms,

the longing to rest – and the quiet strength that carried them forward,

The eagerness to see the playhouse completed.

They wrote of arguments that flared,

differences that softened and gave way,

and the tender thread of togetherness woven through it all!

Language *(English)*: Anther exercise is to translate what they write in their mother tongues into English. At Puvidham, the children translated their own sentences into English and the facilitator helped them to make it into a comprehensive journal in English.

Aligning with centre to make semicircular domes on both sides

Circular window using tyre and making the lower part a planter

Arched door

Dome completed

for construction. This was done to improve insect and termite resistance quality of the mud.

Once the marking was done, and the mud, sand, and water ratio determined, the whole school got involved in the project. Everyone worked for two hours after school everyday to transport stones for the foundation and bricks and mortar for the walls. The 7th and 8th level and 5th and 6th level kids took the lead and they became the masons; all the others were unskilled labour. Some were supplying drinking water or making juice and providing snacks.

In 20 days, the basic structure was ready. The plastering on the inside was done by the 3rd and 4th level kids and the 7th and 8th kids did the water proofing with lime mortar on the outside.

When it was ready, this wonderful structure became everyone's favourite space.

When children are trusted to follow their inner compass in a non-judgemental space, and performance is not evaluated through exams, they naturally blend freedom with accountability, and this balance leads to meaningful, joyful, and creative learning that remains with them as beautiful memories not by rote but because of the experience and joy.

Meenakshi Umesh started Puvidham Learning Centre, based on Gandhi's Nai Talim after she decided to move to Dharmapuri, a village far away from Mumbai. Her move came after she realized that city life was harming her and her children. She can be reached at <director.meenakshi@puvidham.in>.

Assessment and evaluation: what they mean in the math classroom

Monica Kochar

In our mission as educators, particularly in the realm of mathematics, we constantly strive to understand how well our students are grasping the intricate world of numbers, shapes, and patterns. Two terms that frequently surface in our discussions about student progress are "assessment" and "evaluation." While often used interchangeably, it's crucial for us, as teachers, to understand their distinct roles and how both can be leveraged to create more impactful learning experiences in our middle school math classrooms.

Think of assessment as a continuous journey of gathering information about student learning *during* the instructional process. It's like a doctor regularly checking a patient's vital signs to monitor their health and make necessary adjustments to the treatment plan ("Assessment and Evaluation", n.d.). In our classrooms, assessment involves employing various techniques to understand what our students know, what they can do, and where they might be facing challenges. This feedback loop is invaluable for both us, as teachers, to refine our teaching methodologies, and for our students, to gain clarity about their learning journey.

Evaluation, on the other hand, is akin to a final diagnosis. It involves using the data gathered through assessments to make judgments about student learning at a specific point in time, often culminating in grades or reports ("Assessment and Evaluation", n.d., p. 1). While evaluation plays a vital role in the academic structure, particularly in determining progress to the next level, its focus is primarily on summarizing learning rather than actively guiding its development (Knowly, 2020).

Summative evaluation

Reflecting on our own experiences as students in India, many of us might recall a system heavily reliant on summative evaluations – the unit tests and final examinations that primarily aimed to assign grades and determine promotion ("Rethinking classroom assessment", 2006). While these evaluations provided a snapshot of our performance at a particular moment, they often lacked the immediate, actionable feedback that could have truly propelled our learning forward, especially in subjects where we struggled. The emphasis was often comparative, sometimes even leading to public comparisons that could be disheartening and fail to address individual learning needs (Next School, 2016).

The modern educational landscape calls for a more balanced approach, one that recognizes the power of both assessment *for* learning (formative assessment) and assessment *of* learning (summative assessment).

The power of formative assessment: guiding the learning journey

Formative assessment is like the compass and map for our students' learning journeys. It provides ongoing feedback that helps them understand where they are, where they need to go, and how to get there. It's about embedding assessment activities seamlessly into our daily lessons to gain real-time insights into student understanding.

Here are some practical examples of formative assessment we can readily incorporate into our middle school math classrooms:

- **Quick quizzes and polls:** Short, ungraded quizzes or interactive polls (using online tools or even simple hand signals) at the beginning or during a lesson can quickly gauge students' prior knowledge or immediate comprehension of a concept.
  - Example (Integers): Before introducing the addition of negative integers, a quick poll asking students to define "negative numbers" or solve simple addition problems with positive integers can reveal their foundational understanding.
- **Think-pair-share:** Posing a question or problem, allowing students the time to think, discuss their ideas with their partner, and then share their thoughts with the whole class. This encourages

active participation and reveals different approaches to problem-solving.

- Example (Fractions): After explaining the concept of equivalent fractions, ask students to think of an equivalent fraction for 52, pair them with a classmate to discuss their methods, and then share a few of the generated equivalent fractions with the class.

▶ **Exit tickets:** Short prompts or questions given to students at the end of a lesson to assess their key takeaways or identify any lingering confusion.

- Example (Algebraic expressions): An exit ticket asking students to write down one thing they learned about combining like-terms and one question they still have can provide valuable feedback on the effectiveness of the lesson.

▶ **Observation and questioning:** Carefully observing students as they work on tasks and asking targeted questions to understand their reasoning and problem-solving strategies.

- Example (Geometry): While students are constructing triangles with specific angle measurements, observe their techniques and ask questions like, "Why did you choose that angle?", or "How do you know these two sides will meet?"

▶ **Self and peer assessment:** Providing students with rubrics or clear criteria to evaluate their own work or the work of their peers. This fosters metacognition and a deeper understanding of learning objectives.

- Example (Data handling): After a project on representing data using bar graphs, provide students with a rubric to self-assess their graph based on elements like correct labelling, accurate representation of data, and clear title. They can then use the same rubric to provide constructive feedback to a classmate.

▶ **Two stars and a wish:** A simple feedback strategy where students provide two positive comments (stars) about a piece of work and one suggestion for improvement (wish).

- Example (Mensuration): After solving problems on areas of rectangles, students can review a classmate's solution and offer two things they did well and one area where they could be more precise, perhaps in their calculations or explanations.

Formative assessment is not about assigning marks; it's about providing timely and specific feedback that empowers students to take ownership of their learning and make necessary adjustments (Knowly, 2020). It helps us teachers to identify learning gaps early on and adapt our instruction to address those needs effectively ("Rethinking classroom assessment with purpose in mind", 2006).

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The role of summative evaluation: measuring overall achievement

Summative evaluation, while focused on assessing learning at a specific point, still holds significance in our educational system. It helps us gauge the overall achievement of learning objectives and provides data for reporting student progress. However, it's crucial that summative assessments are designed in a way that truly reflects student understanding and not just rote memorization.

Examples of summative evaluation in middle school math include:

- **Unit tests:** Covering specific topics or units of study. These should ideally include a variety of question types (multiple choice, short answer, problem-solving) to assess different levels of understanding.
  - Example (Ratio and proportion): A unit test on ratio and proportion might include questions on finding equivalent ratios, solving direct and inverse proportion problems, and applying these concepts to real-life scenarios like sharing amounts in a given ratio.
- **Mid-term and annual examinations:** Comprehensive assessments covering a broader range of topics. These should aim to assess not just factual recall, but also the ability to apply concepts and solve more complex problems.
  - Example (Algebra): A midterm exam on algebra might include questions on simplifying expressions, solving linear equations and inequalities, and applying algebraic concepts to word problems.
- **Projects and presentations:** Allowing students to demonstrate their understanding through in-depth exploration of a topic and presenting their findings. This can assess not only mathematical knowledge but also research, communication, and critical thinking skills.
  - Example (Geometry): A project where students research different types of quadrilaterals, their properties, and real-world applications, culminating in a presentation with visual aids and explanations.
- **Performance tasks:** Engaging students in activities that require them to apply their mathematical knowledge and skills to solve realistic problems.
  - Example (Data handling): A performance task where students collect data on a topic of interest (e.g., favourite sports of classmates), organize it in a table, create appropriate graphs, and interpret the findings.

To make summative evaluations more equitable and reflective of true understanding, we can:

- Provide clear rubrics: Sharing the criteria for assessment beforehand helps students understand expectations and focus their learning accordingly.
- Offer choices: Where appropriate, allowing students to choose from different assessment tasks can cater to their strengths and interests.
- Incorporate real-world applications: Designing problems that connect mathematical concepts to real-life situations can make the assessment more meaningful and relevant.

Striving for a balanced approach

The key to effective assessment and evaluation in middle school math classrooms lies in achieving a harmonious balance between formative and summative approaches. By weaving formative assessment strategies into our daily instruction, we can gain continuous insights into student learning and provide timely support. Summative evaluations then serve as meaningful checkpoints to gauge overall progress and achievement.

By shifting our focus from solely relying on marks to embracing assessment as a tool for guiding and enhancing learning, we can create a more supportive and effective environment where every student has the opportunity to truly understand and appreciate the beauty and power of mathematics. This thoughtful and purposeful approach to assessment and evaluation will undoubtedly contribute to more transformative learning experiences for our students.

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Will the new CBSE two-exam model for class X benefit all of India?

S Mundayoor

CBSE has finally confirmed a major change in the class X examination schedule from 2026. The final board exams will be advanced from March, when they are usually held, to February. There will be an optional second exam in May for those who would like to try and better their scores. CBSE had, in the first week of March 2025 sought feedback on this proposal from schools, state governments and individuals, but it appears that there was hardly any discussion on the impact of this proposal in the media or other public forums.

On the face of it, this proposal looks very attractive and seems compact: students are being provided with one more chance to improve their performance, if their first attempt was not satisfactory. The first exam is compulsory for all. Secondly, the proposal aims to complete all CBSE exams in less than a month, from Feb 15 to March 6. The results for class X will be 'available' on April 20 in DigiLocker, but will not be formally 'announced' till June 30. Thirdly, the second examination is optional for students and is proposed to be held from May 5 to May 20. CBSE

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will declare the formal results on June 30. Hence, class XI "admissions and session can commence" from the last week of April itself.

But the question is: Do CBSE schools in all states in India begin their new academic session in April? In Arunachal Pradesh, for instance, where only the CBSE Board exists, all schools – both private and government – open only by June every year. So do the CBSE schools in Kerala, Karnataka, and Tamil Nadu. Even if it were by an order of the Union Ministry of Education, would it be fair to ask the schools that have just started their academic sessions by mid June 2025 to curtail their class X sessions and get ready for the final exams by February 2026? Can one cot fit all?

For decades, the CBSE class X and XII exams were being held from March to mid April. This meant that school sessions and activities in January and February were practically undisturbed by exam schedules. Nor did the CBSE or school-level exams clash with the festivals of Arunachal Pradesh, most of which fall in February. Now, according to this new timetable, a major exam for class XII will have to be written on February 20 – Arunachal Pradesh's Statehood Day!

These were a few disturbing thoughts that emerged when CBSE made this proposal in March 2025. Conducting two exams, one after another in a span of two months: will this benefit or is it suitable for students in Arunachal Pradesh, when 62.6% of them are in government schools (ASER Report 2023), and most of them rural students, staying and studying in schools far away from their homes?

In an article in the *Arunachal Times*, March 6, 2025, (https://tinyurl.com/23z92k3n), I had raised these issues. A submission was also made to the CBSE Chairman on March 9, 2025, explaining the unsuitability of a sudden change in CBSE exam schedules for the government school students of Arunachal Pradesh. I also pointed out that –

1. For most government school students in Arunachal Pradesh, passing the class X exam itself posed several challenges (in 2023-24, the class X results in Arunachal Pradesh was 49%, this year, 56.1%);
2. The proposed second exam in May falls in the agricultural season in Arunachal, when most students from the rural communities are involved in family farming activities. For families from other states serving in Arunachal, May is the only time they get to go to their hometowns.
3. May being a rainy month, Arunachal Pradesh suffers from frequent disruptions in power and internet connectivity, making it difficult to learn and read. The students will also have no mentors to guide them to prepare for the second exam.
4. Since most students study in schools located far away from their villages, staying in these school-towns in May will cause the families, especially of adolescent girls, accommodation and financial burden.
5. Teachers deployed for examination duties in May will go on their vacations between June and August, after schools re-open. This will lead to further academic instability in government schools, where skilled subject teachers are always in shortage.

Hence, the second exam proposed by the CBSE may be of interest only to a section of students in the few Kendriya Vidyalayas (KV) and Jawahar Navodaya Vidyalayas (JNV) (which follow a common academic calendar across India, no matter which state they are in) and the expensive private schools in Arunachal.

Before the new exam timetable is introduced, Arunachal Pradesh will have to re-schedule its academic calendar to begin in April. Such an overnight change is neither feasible nor suitable.

Hence, it will be in the best interest of the student-community, if the CBSE –

i) Initially implements the two-exam scheme on a pilot basis in the metros, KVs, and JNVs, before rolling it out across the country.
ii) Introduces a flexible exam schedule, taking into consideration the local socio-cultural and climatic conditions of different regions in the country. While some states may find February convenient, states like Arunachal will prefer March. Even if a change is desirable, it should give the states adequate time to adjust to the new proposal.

Shouldn't our school academic and examination calendar be planned to suit the climatic and socio-cultural background of each state? Something for our education planners and leaders to think about?

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ECOWATCH

Under the tamarind grove

Nandini Dholepat

Photos courtesy: Nandini Dholepat

After a short interval, the Backyard Discovery Group was becoming active again. Let us see what places the group members are going to explore this time.

I found a message from Sujatha in our Backyard Discovery WhatsApp group. "Nandini, we had a good experience at the university. Our children's annual exams will be over by the end of March. Can you suggest a place to visit before we leave for our hometown?"

"I will think about it and get back to you. In the meanwhile, if someone else also knows, please do suggest," I replied.

I started thinking about places around the city where there wouldn't be much of a crowd, and where we could leisurely spend time and learn something.

The next evening, Sujatha and Madhavi came to my house. They started discussing their children's exams and our activities. Suddenly, Sujatha pointed to the bookshelf and asked, "Nandini, what is that wooden thing on the bookshelf?"

I took the wooden box out and opened it in front of them. "This is *Halaguli mane*. It is a traditional board game. Two people can play the game. We should count and fill each cup with an equal number of tamarind seeds on both sides. The game starts with removing the seeds from one cup and distributing them in successive cups," I explained.

Madhavi added, "We call this *Pallanguzhi*. It is simple but tricky at the same time. We can play many different kinds of games with this. Children can also learn counting. We played with seashells when we were young, as we lived in a coastal town."

"Interesting. Now, where can we get so many seeds? We use tamarind paste for cooking these days," Sujatha lamented.

"We can go to the Nallur Tamarind Grove and collect seeds," I replied with a smile.

Sujatha and Madhavi said together, "What ... tamarind grove? Where is it?"

"It is on the outskirts of the city. We can go for a picnic after the children's exams," I said.

"Oh! I have never heard about it. I will book a vehicle. We can start early in the morning and spend the whole day," Madhavi said enthusiastically.

The picnic day was fixed and everyone exchanged messages on what to carry, where to start from, and how to reach. We started early to enjoy the morning breeze in the hot summer. As we moved out of the city, we observed rose gardens on both sides of the road.

Vivan asked, "How come so many rose gardens here? All different colours of roses are growing."

"Oh! I never expected so many huge gardens. Look here, you can see vineyards too," Shilpa pointed.

"Yes dear, all these crops are cultivated on a large scale. They supply to the city market and also export. Unless we come out of the city once in a while, we will miss these beautiful sights," I said.

We came upon a signboard with an arrow mark that said *Hunasethopu*. The driver turned the vehicle in that direction.

"Is that a signboard for the Tamarind Grove?" Rini asked.

"Yes, tamarind is called *hunase* in Kannada, *imli* in Hindi, and *chintapandu* in Telugu," the driver said.

As we reached the location, the driver parked the vehicle in front of a temple, which was dedicated to the local goddess, Gangamma. Everyone stepped out and gathered near the temple. "Let's walk around this temple before the weather gets too hot. Carry your water bottles and wear your caps," I instructed.

"It is amazing. Here, all the trees are only tamarind trees. I have never seen such a huge area full of tamarind trees," Madhavi said.

"Come here, read this information board to get some idea about this place," I moved towards the board.

Anand started reading loudly, "This grove is in Nallur village, near Devanahalli. The tamarind grove is around 250 acres. It is believed that tamarind trees originated here in the 13th century when the Chola kings ruled this area. From that time, it is considered a sacred grove. In the grove, approximately 300 trees can be found. One tree is around 410 years old. It has been identified through carbon dating. Others may be more than 200 years old. The Government of Karnataka declared this a heritage site under the Biodiversity Act 2002."

"I want to see the oldest tree. Let's run," Vivan shouted and started running. The other children followed.

"These children do not pay attention to anything; in which direction are they running?" Sujatha asked in frustration.

"Not to worry, they will come back, then we will all go together," I said.

As the children returned, breathing heavily, I said, "If you want to see this grove, you must follow me. Walk carefully, the Lantana plants growing on the sides may irritate your skin."

"Do we find snakes here?" Rini asked.

"Of course, you will find many small animals like mongoose, hare, snake, and chameleon. Observe the different kinds of birds and butterflies," I said.

We moved along a narrow walkway. We observed many old and young trees. The children jumped and plucked tamarind fruits from low-hanging branches.

"My mouth is watering just by looking at these fruits. It is tasty to eat unripe fruit with salt." Madhavi said.

"When we just think of gooseberries or tamarind, why does our mouth start to water?" Shilpa asked.

"Because you've already tasted it. The smell, taste, or even the sight of some foods releases saliva to digest it," I said.

"Look at the canopy, it is spread like a big umbrella. See these pink flowers. They are beautiful. I never saw tamarind flowers before," Madhavi said.

"The other tree has very few branches and leaves. Each tree is different," Shilpa pointed.

I called everyone. "All of you come here. This tree is very huge. The bark is gnarled. It is hollow from the centre. But it has green foliage. This is one of the oldest trees."

"It looks so strange. Why is it so knobbly?" Shilpa asked.

"Over the period, it must have experienced many insect infestations and abrasions due to wind, rain, or human acts. But it recovered from the damage and kept growing. That is why it is twisted," I replied.

"I wonder how only tamarind trees are here," Anand said.

"It is a multipurpose tree. It grows well in less rainfed areas, without much maintenance. People who lived here must have planted a few, and later they multiplied," I said.

"Look here what I got! We can play with this today," Anand said, showing the seeds he had collected.

"There is an old temple here. Let us sit there for some time and then leave," Madhavi called everyone. All of us sat on the stone bench to relax.

Suggested activities

1. Make a list of food items that one adds tamarind to.
2. Which other sour fruits are used in cooking?
3. Name the different types of seeds that are used for playing games.

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A mission to improve FLN

Faiz Mohammad Qureshi

Photos courtesy: Faiz Mohammad Qureshi

Foundational literacy and numeracy has been acknowledged as a critical imperative in the National Education Policy (NEP) 2020. Various studies (such as the National Achievement Survey, ASER, Pratibha Parv) and experiences have been indicating the necessity of this for a long time now. Following the introduction of NEP 2020, several states have taken positive initiatives in this direction from the academic year 2021-22. These initiatives include infrastructural arrangements, human resources and understanding, curriculum development, textbooks, teacher guides, workbooks, and capacity building of teachers and functionaries.

To address this need the Madhya Pradesh government launched the *Mission Ankur* programme under the NIPUN Bharat initiative in 2021-22. Here, I give some insights about this well-structured and academically enriched programme.

Programme insights

This programme has been designed to facilitate early literacy and numeracy acquisition. Some of the processes and activities are based on experiential assumptions, while others have been designed according to the specific needs of the programme.

Two key aspects emerge here: first, how children learn, and second, how educators can be guided to implement effective teaching strategies in their classrooms. A critical challenge lies in accurately assessing teachers' capacities and the actual instructional time available in their classrooms. Often, a training session is conducted, followed by

classroom implementation, and then a refresher or subsequent training session. However, the content of these training sessions is typically designed by academicians based on limited observation and experience.

In most cases, the training framework for an entire academic year is predetermined and implemented at fixed intervals, aligning the program's operations with its overarching objectives. To navigate this complexity, real-time data analysis has been integrated into monthly meetings and short-term training sessions. This data-driven approach constitutes a significant strength of this programme.

Availability of materials

Another major feature of the programme is the availability of Teaching Learning Materials (TLM). Ensuring that every school has maximum access to supplementary teaching materials for Hindi, English, and mathematics has been a significant achievement. From an academic perspective, these materials are highly useful and enrich the teaching-learning process.

However, one crucial challenge is ensuring that teachers know how to use these materials effectively. Most teachers are familiar with various teaching methods, and those who need additional guidance are provided training under this programme.

Another concern is that while many teachers know how to use the materials, they struggle to integrate them into their daily lessons or use them as needed – either due to time constraints or classroom management issues. Lastly, the teachers' interest and willingness to use these materials also play a significant role.

Workbooks and teacher handbook: the core of the programme

The most crucial components of this programme are the workbooks and the teacher handbook. For all subjects from grades 1 to 4, every lesson has been structured weekly, with each day's activities clearly described. This is designed to ensure that if the outlined activities are followed, students will learn effectively.

The workbook includes three periodic assessments to assess students' learning levels that help them progress. Each page contains engaging, structured academic exercises. To ensure the effective implementation of these workbooks, the program includes orientation sessions, training, monthly discussions, and a detailed teacher guide that provides step-by-step instructions on how to work on each page.

In addition to this guide, the programme also integrates lesson planning, training on teaching methods, and scaffolding support provided by educational mentors (*Jan Shikshak*). This comprehensive approach enhances the teaching-learning process.

Teacher autonomy

A significant concern with this programme is regarding teacher autonomy and the assumption that *"teachers do not know how to teach."* Chapter 3, Page 560 of the National Curriculum Framework 2023 (NCF-SE 23), however, presents a different perspective on teacher autonomy:
"Teachers must have the pedagogic autonomy to plan and organise content, decide the sequence and methods of teaching students as the situation demands, along with ways of assessing their learning..."

In this context, the extensive structuring of the workbooks appears to be overwhelming for teachers. Teachers are required to ensure that all students complete at least three to four pages per subject per day, review their work, and sign off on it. Managing this alongside other non-academic responsibilities makes the process exhausting. Ultimately, this critical educational activity often gets reduced to mere "filling and completing of workbooks."

Practical challenges in classroom implementation

Another major issue is that students who are irregular in attendance fall behind in completing the workbook pages according to the weekly schedule. In just one or two weeks, the gap between the regular and irregular students widens significantly, forcing those who missed lessons to copy from their classmates. Teachers, too, find themselves facilitating this process because they are required to verify and sign off on each day's work.

Additionally, teacher availability and time management in schools pose serious challenges. The situation varies – some classrooms have 45-50 students, while others have 25 students across multiple grades (multi-grade teaching). In both cases, classroom management and ensuring personalized attention to each child remain a major concern.

Multi-grade teaching is a reality and it was considered while designing the programme's training modules and workbooks. However, training sessions create artificial classroom conditions (for simulation), making it extremely difficult to implement these strategies in actual classrooms. Moreover, teachers are burdened with administrative tasks such as student verification, ID creation, and scholarships, which further divert their attention from teaching.

Despite these challenges, teachers are striving to help their students achieve foundational literacy and numeracy skills. It is expected that students currently in grade 2 will benefit from this programme by the time they reach grade 5. However, for students currently in grade 4, the question remains: *Will teachers continue to use these enriched learning processes and activities even after the closure of the programme by the end of 2027?*

Possible improvements

To make this programme more effective, the following suggestions could be considered:

1. Reduce the length of the workbooks to make them more manageable.
2. Allow teachers some autonomy in their classrooms, so that they can implement teaching strategies flexibly.
3. Convert monthly meetings into bi-monthly sessions and make them entirely teacher-led with a pedagogical approach.
4. Develop a system that helps teachers identify essential instructional practices that might be missing from their classrooms.
5. Exempt teachers who are handling foundational literacy and numeracy from non-academic tasks, so that they can focus solely on teaching.

If we, as a system, can ensure these improvements, we may successfully achieve the goals of foundational literacy and numeracy within the next 3-4 academic years. Moreover, these refinements will lead to richer educational experiences for future generations.

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The messy symphony of learning: rethinking the silent classroom

Tarannum Shaikh

Remember the golden rule of kindergarten, 'sharing is caring'? Apparently, that heartwarming principle gets shoved into a locker when the first-grade bell rings. Suddenly, classrooms morph into hushed zones where teachers become the sole source of knowledge, barking out instructions while students passively absorb information. Gone are the days of collaborative block-building projects and vibrant finger-painting sessions. Now, the mere thought of sharing a snack sends shivers down some teachers' spines. Imagine this scenario: a child, eyes wide, reaches for a bite of their friend's idly. "Don't put your hands in others' tiffins [lunchbox]," the teacher scolds, effectively shutting down the possibility of social interaction and a potential lesson in generosity. Her justification? A fear of "mess" and prioritizing order over social interaction. Another child, eager to express themselves creatively, asks to borrow a crayon. "Why don't you bring your own crayon box?" comes the reply, nipping collaboration in the bud. "No, don't ask others, colour it at home," the teacher insists, fostering a sense of isolation rather than encouraging resourcefulness and turn-taking. These seemingly insignificant exchanges highlight a troubling trend – a classroom environment terrified of a little "chaos."

Why are we so afraid of a little noise and a few crumbs? This rigid approach might seem efficient, with teachers maintaining control and students

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appearing focused. Teachers crave order and a quiet learning environment. This response highlights a crucial point: teachers prioritize order over social interaction. But why? There are several factors at play. Firstly, teachers are often under pressure to cover a specific curriculum within a limited timeframe. Disruptions caused by sharing food or arguing over crayons can eat into valuable instructional time, putting them behind schedule. Secondly, maintaining a controlled environment can feel essential for classroom management. Unstructured interactions, like sharing snacks, can lead to unexpected situations – spills, arguments, or a domino effect of other children wanting to share. While these situations might seem manageable to us, for a teacher juggling a room full of energetic youngsters, the potential for chaos can be daunting. The "no sharing" rule becomes a preventative measure, a way to nip potential disruptions in the bud. Finally, some teachers might simply be less comfortable with fostering a more social learning environment. They might feel more confident delivering direct instruction and may not have the training or experience to effectively navigate the messiness of social interactions within the classroom setting. In their minds, enforcing a "no sharing" rule ensures a smooth flow of instruction, even if it comes at the cost of stifling valuable social interaction. While this approach might achieve a semblance of order in the short term, it could hinder children's social development in the long run. Sharing experiences, even messy ones, allows children to build relationships, practice empathy, and learn conflict resolution skills – all crucial tools for navigating the social world.

Imagine a classroom alive with sound: whispers of crayon trades, negotiations over the last juice box, maybe even a heated debate over the coveted red building block. This "chaos," as some might call it, is actually a goldmine for social development. Research by Kostelnik et al. (2011), tells us why. Social interaction during play is a breeding ground for essential skills – empathy, communication, and conflict resolution. Picture a disagreement over a shared colouring book. Kids learn to express themselves, listen to each other, and ultimately find a solution (sharing maybe or taking turns). This messy process, far from disruptive, is a social learning masterclass. Here's the science behind the mess: research by Pellegrini and Bjorklund (2008) suggests that overly sanitised environments might hinder, not help, development. Play, with its inherent messiness and need for negotiation, allows children to experiment with different roles, resolve conflicts (peacefully or otherwise), and build friendships – all crucial tools for navigating the social world. Sharing a crayon might lead to a squabble over colour, but it also teaches turn-taking and compromise – essential for future success. Adding to the evidence, Darling-Hammond et al. (2009) suggest that fostering collaboration and student interaction can actually lead to deeper learning. In other words, by embracing some playful "disruption", we might be setting our children up for better social and academic success. Yet, the idea of accepting this disruption as part of the learning process can be a hard pill to swallow. Teachers often find themselves shushing and scolding students for talking and sharing.

One funny incident, however, perfectly illustrates the contradiction! After witnessing such corrections day after day, as part of my preschool research on how biases and social attitudes get formed at an early stage, I came across a sight that stopped me in my tracks. One day, I saw all kindergarteners and their teacher on the playground. Have you ever seen a bunch of kindergarteners unleashed on a soccer field for the first time? Forget fancy footwork and intricate plays – for them, it's pure, unadulterated joy. Kicking that round ball around is way more fun than memorizing offside rules or strategizing for victory. They chase it with gleeful abandon, a temporary truce declared as they all reach for the coveted sphere. But here's the kicker (pun intended): imagine their surprise when the teacher, initially excited about some playtime, blows the whistle and shouts, "When will you learn to share?!"

It's a question that leaves you scratching your head. After all, these same kids were told that "no sharing" applies to everything from lunchboxes to crayons. So how, in this chaotic sea of tiny legs and boundless energy, are they supposed to magically develop the art of turn-taking and cooperation? Isn't this a classic case of mixed messages? But there's a disconnect, isn't there? At home, in many upper-middle-class and even lower-middle-class families, children are showered with love and gifts. They often don't experience the need to share because their desires are readily fulfilled. School, too, reinforces this by sometimes prohibiting sharing due to concerns about mess or disruption. So, the question remains: how exactly are we expecting them to learn this crucial social skill?

It's time to rethink the silent classroom. Instead of "no sharing" edicts, what if we created spaces where children learn through interaction? What if, instead of stifling interaction with rigid rules against

talking or sharing, we created spaces where children learn through collaboration and social engagement? Research by Rubin et al. (2010) suggests this very approach holds the key. Their findings show that social play acts as a breeding ground for essential life skills – empathy, communication, and conflict resolution. Unlike situations with high stakes (like a test or presentation), play allows for trial and error without significant consequences. This allows children to develop empathy by taking on different roles and perspectives during play. They practice communication by negotiating rules, expressing desires, and resolving disagreements. Conflict resolution specifically benefits from play as children encounter situations where they need to compromise, share, and find solutions that work for everyone involved. These are precisely the abilities our children need to navigate a world that increasingly demands strong social and emotional intelligence.

This doesn't mean abandoning structure altogether. Boundaries are still important. However, consider the power of student-led learning, an approach championed by John Dewey (1938). Dewey, a prominent American philosopher and educational reformer, believed that learning happens best through hands-on experience within a social context. Imagine a classroom where students collaborate to create their own rules for playtime or take turns cleaning up after lunch (think of the life skills learned!). This "messy" process of working together, as some might perceive it, is actually a chance for them to develop problem-solving abilities and a sense of responsibility. Similarly, taking turns cleaning up after lunch fosters not only practical skills but also builds a sense of community and shared purpose, mirroring Dewey's emphasis on learning within a social context. A social context, as Dewey saw it, is an environment where students interact with each other, share ideas, and grapple with problems together. This collaborative effort in cleaning up after lunch isn't just about wiping tables; it's about students learning to see themselves as part of a larger group with shared goals. They contribute to the wellbeing of the classroom as a whole, fostering a sense of belonging and responsibility. This social engagement is central to Dewey's theory, as he believed it allows students to test their ideas, refine their understanding, and develop essential social skills. By allowing them to take ownership of their classroom environment, we're not just teaching them to clean, we're laying the foundation for them to become active and engaged participants in their communities.

Imagine a classroom transformed. Gone are the rows of silent desks, replaced by a vibrant learning space, where students huddle together on colourful rugs. Sunlight streams through windows as laughter

mingles with the murmur of discussion. In this picture-perfect scene, students are no longer passive recipients of information, but active participants in shaping their learning environment. The teacher sits cross-legged on the floor, not at the head of the class, but alongside her students. Together, they're finalizing a set of classroom guidelines collaboratively created through lively debate and negotiation. One student proposes a "quiet corner" for independent reading, while another suggests a designated space for collaborative projects. The teacher facilitates the discussion, ensuring everyone has a voice and that the rules established are fair and achievable. This student-led approach fosters a sense of ownership and responsibility. The rules aren't imposed from above, but rather a product of their collective effort. As a result, students are more invested in following them. Healthy social interaction becomes the norm, as students navigate disagreements with the tools they've learned through guided discussions. Teacher remains a constant presence, not as a disciplinarian, but as a supportive guide, ready to offer gentle nudges and encouragement when needed. This is the essence of a student-centered classroom – a place where learning flourishes through collaboration, social engagement, and a shared sense of purpose.

So, let's embrace a little "creative chaos." It might just be the secret weapon our children need to thrive in a world that demands strong social and emotional intelligence. By fostering interaction and collaboration, we can transform classrooms from silent rooms into vibrant social hubs, equipping our children with the skills they need to not only succeed academically, but also become well-rounded individuals who can navigate the complexities of the future.

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Transforming a reluctant reader in six days

Neeraja Raghavan

Have you struggled to get your child/student to read? Especially in today's WhatsApp era, where everything that needs to be said is to be done in as few words as possible?

If you have, you might enjoy this true story. Better still, you may find that this anecdote can initiate a brainstorming session in your own school.

If you can use this episode in some way in your own working environment, then the purpose of this column will be fulfilled!

It was during the COVID pandemic that the principal of a school noticed something on her virtual rounds. One boy was consistent in his resistance to read: he always claimed to have an internet connection with very poor bandwidth. [He didn't seem to have a problem with his internet when he was *not* called out to read, however.]

The principal spoke to the boy concerned and requested him to send her an audio recording daily by 5 pm, of him reading a short extract aloud. Barring an initial tardiness, the recordings began dropping into her WhatsApp daily and voila! *within six days*, she noticed a significant improvement in his vocabulary as well as his self-confidence. When the principal shared the read-aloud recording with the boy's English teacher, asking her to guess who this boy was, the teacher was unable to do so!

Clearly, we teachers miss the hidden potential of some of our students – human as we are!

The Principal then reflected on the root cause for this speedy success.

In a half hour clip (please see the link at the end of this article), she describes the nature of her **instructional intervention**. Aimed at a specific change, an instructional intervention can be formalized – and it will often demand follow-up. In this case, the principal systematically followed up with the student for about a week, and was very satisfied with the results.

There is more to this story. When this principal shared her experience with her colleagues who were all part of an online group, they reflected collectively on the possible reasons for her success. The following questions were discussed:

- Why did this technique work?
- When will this strategy NOT work?
- What are the obstacles to using instructional intervention?
- What are the enablers that you can identify?
- What are some examples of problems that can be tackled using instructional intervention?
- Can you recollect any such situation, where you have unknowingly intervened, but could not follow up?
- Do we always need to involve parents in such interventions?
- How can we adapt the same to a larger group?

Discussion points in the online group of teachers.

[In order to get a full picture, you will need to watch this clip. I am capturing the essence of the discussion here.]

How did this transformation happen so quickly? The principal shared how she often conveys to students that they need to alter their attitude – if she feels that is needed. What enabled the principal to get such a simple idea and *why didn't it strike the teacher concerned*? When – if at all – does a harried teacher get ideas like the principal got, to tackle such a problem? The importance of loosening one's grip on a problem and brainstorming with colleagues emerged during this online discussion. As for the time that a teacher (imagines she) needs to invest with students individually, the group reflected on whether or not a huge investment of time was truly needed.

Further, during the online class, as the boy's father passed by, he heard his son's name being mentioned several times. This triggered his interest, and quickly drew him into the practice of getting his son to read with interest. This elicited the question from the group: *Do we always need to involve parents in such interventions?* What if the parents are themselves not literate and the child is a first-generation learner? Will such support even be possible then?

The online discussion concluded with a description of how Helen Keller learnt her **first abstract word**: and remember, abstract words are hard enough for a child *with all faculties* to master! Helen Keller was a blind, deaf, and mute child: the account of how *she* learnt abstract words capped the discussion and set the teachers thinking.

Do write in and let us know if this month's story set you thinking, too!

Watch the half-hour clip here: https://www.youtube.com/watch?v=hftOrj4abOs&t=62s

Neeraja Raghavan is founder-director of Thinking Teacher (www.thinkingteacher.in), an organization that networks with teachers across the country. Thinking Teacher aims to awaken and nurture the reflective practitioner within each teacher. Thinking Teacher's goal is to help build deep inquiry and rich learning into the teaching process. The author can be reached at <neeraja@thinkingteacher.in>.

Thinking about thinking

Timira

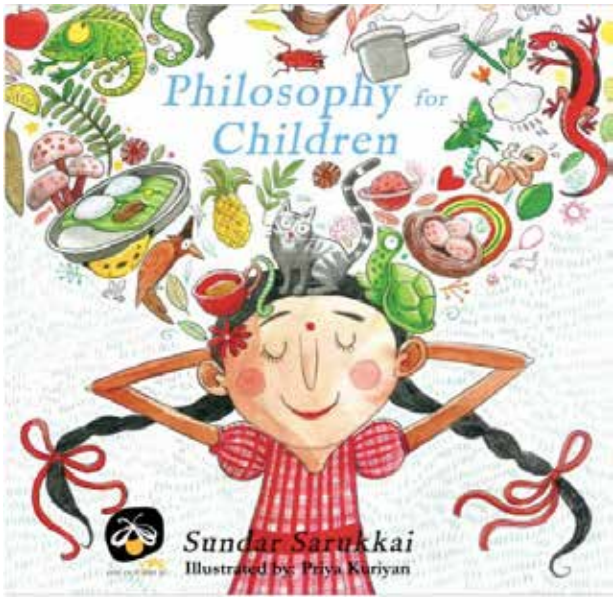
Teachers from DPS Pune exploring the *nature* of questions.

Photos courtesy: Timira

How does learning happen? We may say it happens through discovery, exploration, experimentation, through exposure to information or experience. There are several ways. However, at the core of it, only when we think and reflect about it does learning happen. Learning is a consequence of thinking. This brings me to the fascinating topic of thinking about thinking, which we almost never think about!

Where does thinking happen? Why is thinking important? Do we think all the time? Do we all think in the same way? Are there different ways of thinking? In what language do we think? Can we think without a language? Do creatures other than humans think? If they do, how do they think? Can we improve the way we think?

I'm going to borrow parts of a brilliant book, *Philosophy for Children*, written by philosopher and writer, Professor Sundar Sarukkai, which has a chapter on 'thinking' that beautifully breaks down this process for us.



Courtesy: Jugnoo Prakashan, an imprint of Ektara.

Let's begin with the most basic question that Sarukkai asks – **what do you do when you think?** If you have to explain what you do when you are eating, it's easy. You take some food, put it in your mouth, then bite and chew, and finally swallow it. Similarly, you can explain step-by-step what you do when you run, jump, cycle, etc.

Take a moment and try to describe what it is that you do when you think!
It's difficult, isn't it! Sarukkai encourages the reader to note down what happens to your body when you are thinking. He asks if there is a place in your body where you do your thinking. We may have been taught that we think in our mind or brain, but can you *feel* your brain thinking just like you can feel that your mouth and throat are cold when you are eating ice-cream? Is it possible that we can also think in our heart? Or in our stomach – after all it does rumble when we are hungry and act funny just before the math exam! Sarukkai provokes the reader further by asking several questions like – when you run, your body sweats; when you think, does your body sweat? Do you feel tired? What part of your body becomes active when you are thinking? We can say that thinking is talking or arguing with yourself. But for a cricketer, it's the body that thinks when a ball is coming their way – they have no time to talk to themselves in that moment!

There are endless ways humans think and the better we know how we think, the better we learn. This curiosity to understand HOW we think has led several educators and researchers to design tools to make thinking visible. *Project Zero* by Harvard University is one such initiative where they designed several Thinking Routines to create cultures of thinking within schools.

A social studies teacher in class 2 teaching a unit on 'types of houses' recently shared this image (see below) in her class and used a popular thinking routine called **See-Thinking-Wonder**. She first asked the

Image generated using AI.

children to state **'What it is they SEE' in the image**. Students threw up answers like:
'a city with skyscrapers and *jhopdis*'
'an old city and a new city'
'different kinds of houses and buildings'
'brick houses with roofs and tall glass buildings'
'colourful houses and blue and grey buildings'

This was a great entry point into the topic. As the image reflected a stark difference in the types of houses, she then asked the children **'Why do you THINK there are such different kinds of houses in a city?'** The responses of students as young as these were absolutely intriguing.
'...because rich people live in tall buildings and poor people live in the small ones.'
Teacher: "What makes you say that?"
'...because tall buildings are fancy and expensive. It doesn't take a lot of money to make small houses like in the picture.'

Another response was:
'...we need different types of houses now because tall buildings have more flats.'
Teacher: "...and why do we need more flats? What makes you say that?"
'...because there are so many more people living in the city than before.'
Another child added in:
'...yes, my mother always says that when she was my age there was not so much traffic, so many cars, so many people in Mumbai.'

The teacher layered the conversation with an additional thinking routine ***'What makes you say that?'*** allowing the learners to make the thoughts behind their thinking visible. The discussion ranged from how the city was changing to socio-economic realities, also making room for personal experiences to come in. Many students talked about re-development of their own building – a dinner-table conversation in every home in a city like Mumbai.

The teacher decided to end the class by giving the third part of the thinking routine as homework. She asked the learners to write a list of questions starting with: **"I WONDER who/why/what/which/where/how..."** using any of these question starters. This allowed absolute freedom of one's curiosity to know beyond what is obvious and students were so excited to do this homework, many of them began right away even before the class was over.

Of course, the teacher would find her way to stilt houses, igloos, and mud huts and all the other things mentioned in the syllabus, but to use this opportunity to get learners thinking ensured they were invested on this learning journey. There are several such thinking routines (*think-pair-share, think-puzzle-explore, circle of viewpoints–a resource link is provided at the end of the article*), each with their own objectives and provocations but all with the single aim to encourage students to be curious and for them to see and know what and how they are thinking about things.

Unfortunately, we don't draw much attention to the quality of questions we ask in the class. The questions given at the end of a textbook chapter are usually recall-based, closed questions and a few reasoning questions, which the learner is expected to answer with reference to the given text. None of these encourage you to *think*. And to top it all is our new friend AI, which almost encourages you *not* to think!

Over the last few years, I have been encouraging teachers to identify the 'nature' of the questions before crafting them in order to make conscious efforts to bring about deeper thinking. It requires a little bit of extra work, in an already very busy life of a teacher, to create a set of 'thinking questions' for every lesson plan. Take a look at the shift in the nature of questions in a grade 5 and a nursery class and you'll get an idea of the impact it can have on thinking. The quality of learning in the class improved dramatically as teachers became conscious

Grade 5 – EVS – Natural Disasters
2-3 existing questions from the lesson plan: 1. What is a disaster? 2. What are the different types and causes of disaster? 3. How can you help during a disaster? 4. What are the effects of an earthquake on the common man's life? 5. What are the problems they have to face?
Write down the list of Thinking Questions you created for this lesson: 1. Have you ever heard of a natural disaster before? What are some natural disasters that you know about? How can people prepare for these types of disasters in advance? (Funnel questions) 2. Why do you think some people may not evacuate when there's a natural disaster approaching? (Probing question) 3. What emotions do you think people experience when they lose their homes in a natural disaster? (Affective question) 4. How are floods and droughts similar, and how are they different? (Comparison question)

Nursery – EVS – How seeds grow
2-3 existing questions from the lesson plan: 1. Can you name something that helps a seed grow into a plant? 2. What does a seed grow into after it starts sprouting?
Write down the list of Thinking Questions you created for this lesson: 1. If you could plant a magic seed, what kind of plant would grow from it? 2. What do you think would happen if a seed was planted on the moon? Would it grow? 3. What do you think a seed would say if it could talk while it was growing underground? 4. Can you pretend to be a little seed? What do you need to grow big and strong?

Extracted from a teacher's lesson plan.

Teachers from DPS Patna engaged in a Socratic Seminar.

of the kinds of questions students were asking. After the initial rigour of consciously wading through this new competency that was thrown at them, it soon became an effortless way of being. They couldn't ask a closed, recall-based question without noticing it anymore!

Bringing about cultures of thinking in schools is the need of the hour and it is essential that teachers themselves internalize this culture first. As part of a programme I co-design, we introduced Socratic Seminars with teachers so they can experience how exciting and enriching a collaborative thinking process can be. A Socratic Seminar is a structured discussion method, where participants collaboratively explore a piece of text through thoughtful questioning and dialogue. It's not a debate, but rather a process of collective meaning-making where participants analyze, interpret, and build upon each other's ideas.

A teacher shared this after their first Socratic Seminar experience: "For the very first time ever I participated and engaged in a Socratic Seminar. It gave me an opportunity to understand and learn how to analyze complex texts and questions.... fostering my own critical thinking and to engage in meaningful dialogue.....which in turn prompted me on my own questioning techniques." Another teacher said: "When we get into discussions, we are always arguing, agreeing, and disagreeing on viewpoints. This was the first time I felt all of us were interested in figuring out together what the article was trying to say. The format of the seminar pushes you to be a better listener and build on other's ideas instead of countering them. Made me think of discussions in a new light!"

Learning as a collective endeavour in an educational setting is ideal. It allows for participants, both adults and children, to actively engage in discussion, critique, and refinement of ideas and thought. If learning is a consequence of thinking, then perhaps we can begin to see thinking too as a collective enterprise. A tiny big shift in our classrooms – ready to give it a shot! What do you think?

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How flexible is our Constitution?

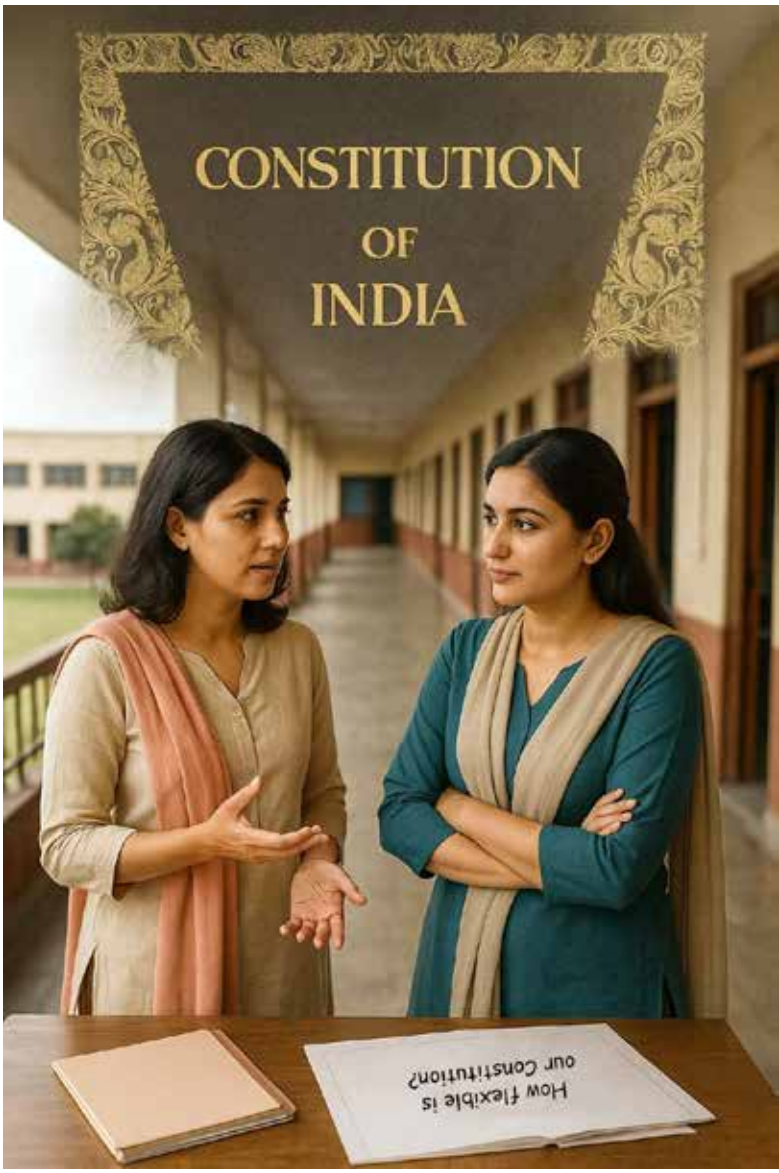
Prakash Iyer

Ira: What would the opposite of federalism be? I can imagine what the structure of the government might be in a small country that does not have many provinces or states, and so a single government would be enough. I am trying to imagine how such a governance structure would look like in India. Did the Constituent Assembly consider this possibility? Why did they consider a federal constitution?

Anubhuti: The word you are looking for is unitary – such a constitution would suggest the kind of government you are thinking of. Yes, this did come up during numerous debates, but the Assembly did what suits our nation and its history, rather than consider one or the other categories of constitutions. In a final address to the assembly, President Dr. Rajendra Prasad said, "We are not bound to have a constitution which completely and fully falls in line with known categories of constitutions in the world. We have to take certain facts of history in our own country and the Constitution has not, to an inconsiderable extent, been influenced by such realities as facts of history."[1] It seems he was saying that we could call it a federal constitution in retrospect after the drafting of the Constitution rather than presume federalism from the beginning.

Ira: This reference to historical facts which have been considered in drafting the Constitution is interesting. Does that mean that as situations change, the Constitution will also have to be modified? That sounds paradoxical! The Constitution would have to say that it is final such that it is adopted by everyone, and at the same time also say that it can be changed. How can they imagine what kind of changes are allowed and what are not?

Anubhuti: Brajeshwar Prasad, a member of the Constituent Assembly said the same thing during the debates. Referring to its federal nature, he said, "It is ordinary commonsense that should tell us that if



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1. https://www.constitutionofindia.net/debates/26-nov-1949/#130353

we want to abolish landlordism you cannot seek the consent of the landlord. ... The purpose of amending a constitution, in effect, will be to take more powers from the hands of the State Governments and confer them on the Centre."[2]

If we give that power to the central government, it does amount to being a unitary constitution. Prasad saw this problem and said if a conflict between provincial and central governments should arise, the Constitution should suggest a referendum – put any controversial amendment to vote to the entire population. He gave convincing reasons justifying a referendum – it would be an appeal to the people and emphasize their sovereignty; it would control the representative governments, cure defects in the party system of democracy, and *curb the absolutism of a party that possesses a parliamentary majority*.[2]

Ira: This would have complicated the issue because they would either have to (a) give the centre more power and veer towards a unitary constitution or (b) adopt an expensive process of voting for amendments.

How did they decide on the amendment process then? There must have been some principles that drove the discussion further. I know we did not choose a referendum because there have been 106 amendments to the Constitution in the last 75 years, but no referendum.

Anubhuti: I read some parts of the Constituent Assembly debates when this discussion happened, and the alternative was also suggested by Brajeshwar Prasad. He said that we could insert a rule that for any amendment, there should be two-thirds majority agreement in both the Lok Sabha (lower house) and the Rajya Sabha (upper house). His argument was that members of the Rajya Sabha will not want to give more power to the central government since they represent federalism and bring in opinions from all states.

He further emphasized that the Constitution should be made *flexible* and not *rigid*. His argument was that the Assembly's vision was *blurred by the effect of the partition* and so they were not able to take an objective view of the needs of a developing economy.

H.V. Kamath, on the other hand, began his response with a fundamental issue – the need to define what an amendment meant. He was concerned that 'amendment' might be interpreted as variation, or even repeal of the Constitution itself.

2. https://www.constitutionofindia.net/debates/17-sep-1949/#123070

Ira: That would be disconcerting. Is amendment defined in our Constitution?

Anubhuti: Yes, it is. Article 368 of the Constitution says, "Notwithstanding anything in this Constitution, Parliament may in exercise of its constituent power amend by way of *addition, variation or repeal any provision of this Constitution* in accordance with the procedure laid down in this article." See, the text solves the paradox you mentioned earlier – how does a constitution call itself flexible and amendable. It defines 'amendment' as changing or withdrawing any *particular* provision in the Constitution, rather than the Constitution itself.

H. V. Kamath further explained two critical reasons for making the Constitution flexible through an easier process of amendment. He said that though the Constitution is a product of representatives of people across India, and they were all wise and educated, they should not associate any superiority to the Constitution they drafted. The reason he gave was that in spite of all this, they were indirectly elected and that too on a restricted franchise. Future parliaments, on the other hand, would be elected on universal adult franchise and so would be *better* in that they would truly represent the entire population.

Secondly, he said, we should allow for the fact that society would progress with time, and "If the Constitution holds up, blocks, the future progress of our country" by disallowing any amendment or making the procedure difficult, "(If it does not bend), people will break it," which was not desirable.[3]

Ira: Isn't it remarkable how our Constitution remains safe and federal, even though it is 75 years now and 106 amendments have been made? Madhav Khosla's point about the Constitution being a pedagogical tool that will teach us democracy seems to be working.

Anubhuti: Yes that is true. Actually, Mr. Kamath also speculated on the issue of unitary versus federal nature of the Constitution through a nice metaphor from physics. He said, "We cannot always guarantee that the unifying forces in the country – the centripetal forces – will gain ground against the centrifugal or the disruptive forces in our land."[4]

To deal with this issue, his suggestion was that if the Lok Sabha voted for an amendment to the Constitution, which the Rajya Sabha rejected, it

3. https://www.constitutionofindia.net/debates/17-sep-1949/#123115
4. https://www.constitutionofindia.net/debates/17-sep-1949/#123133

should be voted for again in the Lok Sabha – this time along with reasons for rejection by the Rajya Sabha. If the resolution wins votes the second time as well, it should be accepted.

Ira: Hold on. Isn't that problematic? It is as if even though the Rajya Sabha is standing with federalism, the Lok Sabha would have the power to insist on an amendment only because they are a majority. To me, this seems like the Constitution would veer to the unitary side.

Anubhuti: You are right, which is why some people say we are a federal constitution regulated by articles 245 to 255, which allow the centre to exercise control on laws and policies of state governments.

The infamous 42nd amendment made during the emergency – which gives the legislature significantly more power than even the judiciary – had to be ratified by the Rajya Sabha and the President.

Ira: Wow! That is disturbing – closest we seem to have come to an armed revolution. By the way, what is the role of the president in all this? We do learn that the president is the chief of the State. Wouldn't the president be able to refuse an amendment?

Anubhuti: That is a tricky question. The role of the president (and governors) is not political per se. He/she is meant to be safeguarding the Constitution, but under the advice of the legislature. The role is more administrative than political.

The judiciary – Supreme Court – has the power to reject any amendment to the Constitution.

Ira: What could the grounds for rejection be? Like Mr. Kamath said, what if at some point both the Lok Sabha and Rajya Sabha vote for changes to the Constitution that makes it unitary or less federal? I can imagine removal of processes and some powers from the states.

Anubhuti: There is something called the Basic Structure Doctrine in constitutional law, which the judiciary has the power to use. According to this doctrine, the legislature cannot change or destroy some fundamental features of the Constitution. These features would vary according to the Constitution. Remember, in the case for reservation for economically weaker sections, two judges argued that it violated the basic structure of the Constitution. Of course, three other judges on the bench said it did not.[5]

The basic structure could be interpreted differently in situations like this, but in simple terms it emphasizes that no amendment can go against the supremacy of the Constitution, the republican and democratic forms of government, secularism, separation of powers, federalism, and the rule of law.

We do know what republican and democratic forms mean – that all citizens participate directly or indirectly in policies and other governmental decisions – which takes care of fundamental rights and duties, and the directive principles of state policy.

The clause about the supremacy of the Constitution is interesting. This doctrine avers that we will remain a constitutional democracy – controlled by the Constitution, not the legislature or the executive. In principle, it ensures that even the government elected by a majority cannot say their party's ideology is superior to the democratic principles defined in the Constitution.

Ira: You mean the social revolution that Mr. Kamath was worried about cannot happen in India?

Anubhuti: Yes and no. You see, somehow by exerting power, the central government did do this during the emergency in 1975. But it was revoked soon after the next elections in 1977, when another party came to power.

Ira: Aah! So, there was no choice about going for elections, in which all of us firmly rejected the party. You would say republican democracy prevailed!

Anubhuti: Yes, it did. We keep learning from our collective experiences with democratic processes; and as I always like to say, democracy is constantly a work in progress.

Ira: I knew you would say this! Also like you said, the Constitution is a palimpsest. Every Supreme Court judgment that interprets articles of the Constitution, overlays it with new meaning; and we learn how to be more democratic than we already are.

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5. https://www.scobserver.in/cases/janhit-abhiyan-union-of-india-ews-reservation-case-background/

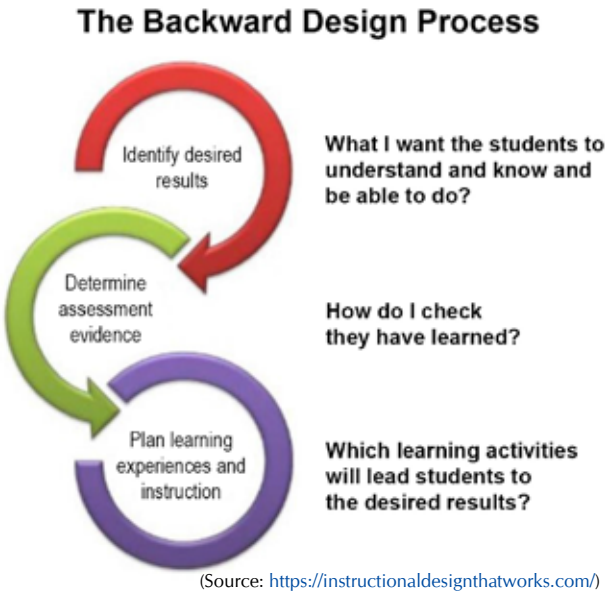
Understanding by design

A framework for effective education

Roopa Vinayak Ram

Social media has crystallized into an all-important public utility – this is what writer and journalist Johnann Hari says in his book, *Stolen Focus-Why you can't pay attention and deeply think again*. Instead of receiving meaningful information, we are often overwhelmed by an endless onslaught of notifications – a steady trickle of 'behavioural cocaine' – telling us of the number of likes or comments received, or upcoming celebrations, and other candy floss, all of which impair our competency to focus and think clearly.

Teaching screenagers has become an arduous task for teachers today. Yes, you read that right, screenagers, that's the term that rightly describes students who are raised on smartphones and social media. These digital natives have become seasoned scrollers and loyal snackers of bite-sized content such that the onus is on teachers to innovate and hook their attention, converting complex concepts into immersive experiences. But a critical question lingers: what can teachers do to build a culture of hard work to assimilate and achieve mastery in subjects, especially when the matter is about students who are primed for instant gratification and snackable knowledge?



(Source: https://instructionaldesignthatworks.com/)

Is there any solution?

This persistent challenge that teachers face can be addressed by an educational framework called Understanding by Design (UbD) conceived by Grant Wiggins and Jay McTighe, two widely recognized educators and authors from USA. Wiggins and McTighe have spent a major part of their careers advocating meaningful educational reforms. In this article, I attempt to illustrate how the UbD model can be used to teach students so that those who struggle with a subject start to understand it. I will be using examples from accountancy to unravel the framework; but, this strategy can be applied to any subject and is therefore useful to all teachers.

The UbD framework: an overview

UbD at its core, is a curriculum design framework that emphasizes *backward design*. In contrast to conventional pedagogical planning, where the focus is on starting with activities or lessons, UbD begins with teachers identifying the end learning outcome and then working backwards to create lessons, activities, and assessments that support those outcomes. In other words, UbD is all about visualizing the big picture and then taking steps to achieve that big picture.

As Stephen R. Covey in his book, *The Seven Habits of Highly Effective People*, rightly pointed out "To begin with the end in mind means to start with a clear understanding of your destination. It means to know where you are going so that you better understand where you are now so that the steps you take are always in the right direction."

What does the framework look like?

The framework consists of three stages:

1. **Identify desired results:** What should students know, understand, and be able to do by the end of the unit?

2. **Determine acceptable evidence:** How do I assess whether students have achieved those outcomes?
3. **Plan learning experiences and instruction:** What activities, resources, and teaching strategies will help students meet their learning goals?

How can UbD be implemented by accounting teachers?

Step 1: Identify the desired results in accounting

- Understanding the fundamental accounting equation (Assets = liabilities + owner's equity).
- Applying the concept of double-entry bookkeeping.
- Knowing how to record basic transactions in a journal.
- Posting transactions so recorded into appropriate ledgers.
- Converting these ledgers into a Trial Balance, which is the first step to preparing financial statements.
- And finally, preparing and interpreting these financial statements (income statement, balance sheet, cash flow statement) to come to a meaningful conclusion regarding the profitability and financial health of the given business.

At this stage, teachers should identify what they want their students to *understand*, not just *know*. For example, students should not only be able to create a balance sheet, but should also understand *why* the balance sheet is structured the way it is and how it provides useful information to business owners, investors, and other stakeholders.

Step 2: Determine acceptable evidence

This is the stage where teachers decide how they will assess their students to check how well they have understood the concepts taught. The conventional assessment methods such as quizzes or standardized exams might not reveal deep understanding. It is here that the UbD framework comes in handy. UbD urges teachers to use *performance tasks* that require students to apply their knowledge in real-world contexts.

For example, one method of assessment could be asking students to prepare financial statements when presented with a set of business transactions. To make the assessment more meaningful, the teacher could ask the students to explain their reasoning for each entry and the implications for the business's financial health. This demonstrates not only technical proficiency, but also a deeper understanding of how accounting information is used in decision-making.

Stage 3: Plan learning experiences and instruction

Once the desired outcomes and assessments have been established, the teacher can move on to planning the actual lessons and activities. This is where accounting, as a hands-on subject, can really benefit from the UbD framework. By focusing on engaging real-world applications, teachers can make accounting more accessible and interesting for students.

Applying UbD to teaching accounting: practical examples

Now that we have a basic understanding of how UbD works, let's explore specific ways in which the framework can be applied to teach accounting.

Image generated using AI.

Example 1: Understanding the accounting equation
One of the first concepts students learn in accounting is the accounting equation: Assets = liabilities + owner's equity. While this equation sounds direct and simple, its inferences are not always immediately obvious to beginners.

UbD application
Desired outcome: Students should not only know the constituents of the accounting equation but also understand its implications in the context of a business.
Assessment: Have students analyze a small business's financial situation. Give them a case study with a set of transactions where they have to identify the accounts involved, classify them into assets, liabilities, or equity. Once this is done, they should also explain how changes in one part of the equation affect the other.
Learning activity: A hands-on simulation exercise where students role-play as managers of a fictional company. Each student is given a role (e.g., inventory manager, sales manager) and must record transactions related to their role. As a group, they will then prepare a balance sheet based on their entries, discussing how each transaction affects the overall equation.

A simulation activity such as this not only reinforces the concept of the accounting equation but also makes the process tangible for students. Learning by doing helps students strengthen their understanding of theoretical concepts.

Example 2: Double-entry bookkeeping
Double-entry bookkeeping is another core concept in accounting, but it can appear complex to beginners. The concept of every transaction having two sides or entries – one on the debit side and one on the credit side – can feel conflicting at first.

UbD application
Desired outcome: Students should be able to not just explain why the double-entry system ensures accuracy in financial records but also explicitly record transactions using the double-entry system.
Assessment: Ask students to complete a set of transactions for a small business, making both debit and credit entries for each transaction. They must also provide a brief written explanation of how the double-entry system prevents errors in accounting.
Learning activity: Use a hands-on, game-based approach where students participate in a "transaction relay." In this activity, one student reads out a transaction (e.g., "The company buys an office computer for ₹ 50,000 on credit"), and the rest of the class sprints to record the correct debit and credit entries. Points are awarded for accuracy and speed, and the class discusses any missteps taken along the way.

When bookkeeping is turned into a game, it is easier to keep students engaged. By repeating the same game with different types of transactions, the rules of double-entry system of bookkeeping are strengthened.

Making accounting easy for beginners through UbD

By using the UbD framework, accounting teachers can customize their instructions to help students learn the basics of accounting and develop their ability to think critically and apply their knowledge in real-world contexts. Here are some additional strategies to make accounting easier for beginners:

- **Incorporate visuals:** Diagrams, flowcharts, and tables can help students understand the relationships between different accounting concepts, such as how transactions flow through the accounting cycle.
- **Team-based learning:** Group activities and projects encourage peer learning and help students develop a deeper understanding of the material through discussion and collaboration.
- **Constant feedback:** Regular formative assessments can help students identify areas where they need to improve before they move on to more complex topics.
- **Everyday examples:** Incorporating real-world case studies, guest speakers from the accounting industry or even fieldtrips to local businesses can help students see the relevance of accounting in the world around them.

Conclusion

Throughout their book, *Understanding by Design*, McTighe and Wiggins present abundant examples, workable strategies, and blueprints to guide educators in implementing the Understanding by Design approach. The complexities of teachers' work and the related oversights that can shroud their minds are plenty, and a framework such as UbD can come in handy. Through interesting, real-world activities, timely feedback, and group activities, learning can become both navigable and pleasant for students.

Roopa Vinayak Ram is committed to creating productive learning environments in schools. She is an accounts teacher at National Academy For Learning, Bengaluru. She can be reached at <roopa304@yahoo.com>.

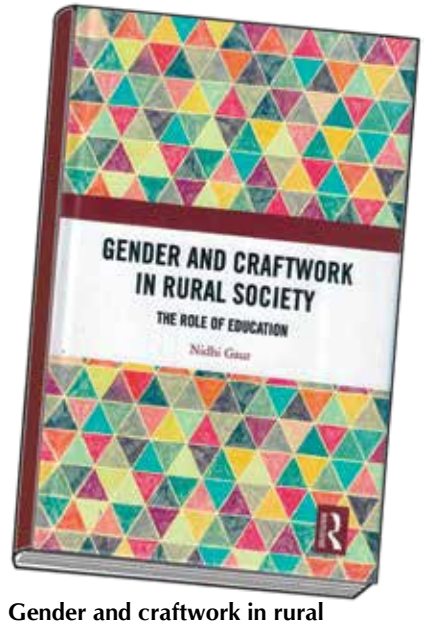
Is craft-centred education the way to go?

Nimesh Ved

This book is an outcome of the author, Nidhi Gaur's, study of Anand Niketan in Wardha, Maharashtra. This is a school where "Work is not associated with particular people; rather, it is perceived simply as work that needs to be done." The first chapter, *Introduction*, deftly conveys what the book is all about; it states, "So far, craft-centred education has not been studied from the perspective of gender, so this study revisits an older idea with a new perspective." The book has been neatly structured over nine chapters spread across 181 pages. The final chapter, *Implications and conclusion*, succinctly sums up the learning – "The examination of the pedagogy of crafts has shown the potential of craft-centred education in the holistic development of the self of the students."

The author "examines the interrelation of gender and the pedagogy of crafts" in depth. She looks at the history and trajectory of crafts and gender juxtaposed with education. She is immersed in the topic and gleans out knowledge from the books she referred to and the children (at Anand Niketan) she interacted with. These she deftly puts forth in the book. She also pauses to clarify terms and concepts (such as modernity and modernization) where the text warrants. She talks about "rural" India, but is aware of the irony that today "the village is referred to as a thing of the past, far away from the 'active' space". The oeuvre of references is rich as well; she brings in John Galsworthy's classic short story, *Quality*, for example, to highlight a point on craftsmanship. She underscores that "craft-centred education has the potential to create a culture of patience, reflective action, expression, coexistence and peace" in a manner befitting the topic.

The book challenges the existing paradigm of school education. In telling us what a craft-centred education can achieve, it points to what we are missing in our conventional schools today. When, for example, the author says of Anand Niketan, "The school was also found to be emerging as a space where students could express and experiment with their gendered roles to understand other people's responses and behaviours towards their experimentation," we go back to the schools we have been associated with as students or teachers and wonder how we had explored gender and what could have been done differently. When a child tells her, "We are taught to study while we learn craft by doing it with our hands," or when she observes that, "The practice of crafts is based on the methods of science," we pause and absorb. This is where the book scores; it nudges us to think. The book also has lessons for today – it informs us that there were critiques on a craft-centred education policy, a little less than a century ago; in other words, education policy matters used to be read and discussed! It reminds us that to move forward as a society, we need to get education into the public consciousness and discourses.



Gender and craftwork in rural society: The role of education
Author: Nidhi Gaur
Published by: Routledge
Pages: 181
Price: Rs.1295/-
Where it is available: https://www.amazon.in/

The editing could have been tighter. Some things are repeated (e.g.: Devi Prasad's introduction and work) and there are errors too (e.g.: it is Zakir Hussain on one page (44) and Zakir Husain on the next (45)). Cliches like "her ngo" too could have been avoided. Could it also have looked at Gandhi with a more nuanced and balanced lens?

The book, however, like all good books, left me with questions:

- Why talk about this school at all when it has neither flourished nor its model replicated?
- Why discuss craft-centred education with an English reading-writing audience; an audience whose aesthetics are not exactly thriving?
- What motivates the teachers in this school? How were their roles envisaged in the initial design?
- Teacher capacity has been a bottleneck in conventional schools; how different is it here?
- Are there other schools, today, that adhere to a craft-centred curriculum? Is a craft-centred curriculum relevant only to the rural areas?
- How is craft education being imparted in our schools today? Has it become restricted to the privileged (read fancy) schools?
- Finally, what if we replace crafts with sports? How would it impact the attributes the author ascribes to craft-based education – gender equity, working with hands, quality, balance, initiative, interdependence?

Nimesh enjoys being amidst books and children. He blogs at https://nimesh-ved.blogspot.com/ and can be reached at <nimesh.explore@gmail.com>.

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